

December 7, 2025 09:00 PM GMT

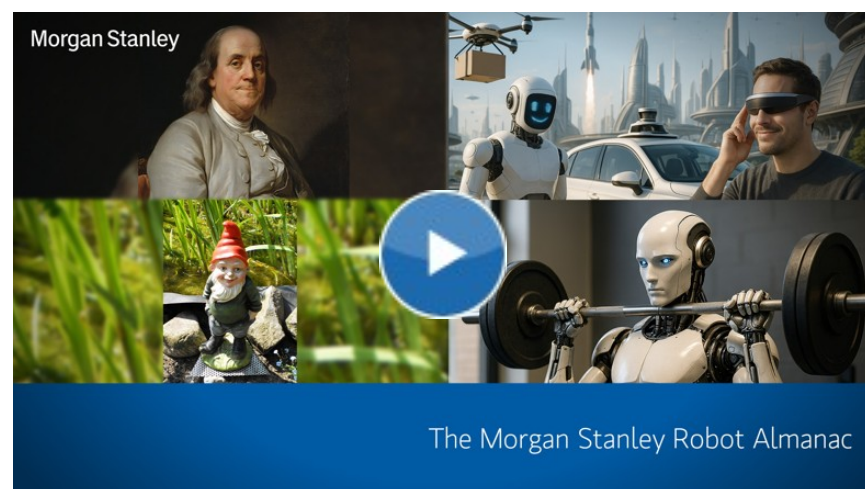
## Physical AI

## The Morgan Stanley Robot Almanac

Our inaugural Robot Almanac is a comprehensive all-in-one guide to the dynamic world of physical AI including our proprietary Robot Model (GROM): a bottom-up forecast of robotics adoption, unit volumes, revenue, and component demand across form factors through 2050.

alphawise 

**Introducing the Morgan Stanley Robot Almanac.** The Morgan Stanley global robotics team has compiled this comprehensive reference guide to help our clients navigate the exciting confluence of AI and the physical world. We believe AI-enabled robotics helps usher in the 3rd Industrial Revolution driving \$25 trillion of combined robot revenues by 2050 with the potential to multiply the \$115 trillion of global GDP over time. This proprietary guide has been compiled, curated, and maintained by the industrials and technology teams across Morgan Stanley's global equity research complex.



**The Almanac is meant to be an ever-evolving, open-source artifact that is organized across a broad range of embodied AI verticals including:**

- Autonomous cars
- Drones
- Humanoids
- VTOLs/Air Transport
- Industrial Robots

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**For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.**

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- Space
- Brain-Computer Interfaces
- Other Form Factors: Warehouse, Agriculture, Construction, Last-Mile Delivery, etc.

**How to use the Almanac.** Following an introductory summary section, the Almanac addresses each primary embodied AI vertical in detail, including:

- Enabling technologies
- Competitive landscape
- Addressable market definitions
- TAM model
- Exposed public companies
- Private company disruptor profiles

The Almanac includes a dynamic table of contents to help navigate the nearly 700 total slides of material in a simple and fun way. This inaugural Almanac will be updated throughout the year and republished on an annual basis to incorporate the latest strategic developments, commercial milestones, technological innovations and regulatory changes to ensure accurateness and relevancy to our clients to help with their own research discovery.

**Due to the size of the Almanac, we have split the deck across 8 'volumes':**

- [Volume 1: AI Gets Physical; Cambrian Explosion of Bots](#)
- [Volume 2: How to Train Your Robot; Geopolitics; Rare Earths Bottleneck; Sagan's Prophecy](#)
- [Volume 3: Humanoids & Industrial Robots](#)
- [Volume 4: Drones & Air Mobility](#)
- [Volume 5: Space & Defense](#)
- [Volume 6: Autonomous Vehicles](#)
- [Volume 7: BCI & Other Form Factors](#)
- [Volume 8: Big Tech's Physical AI Journey](#)

**NEW: Morgan Stanley Global Robot Model (GROM).** As outlined within the Almanac, we introduce the Morgan Stanley Global Robot Model or "GROM" which projects the global robotics TAM (installed base, unit sales, revenue) across key form factors for the US, China, and Rest-of-World. These key form factors include: autonomous vehicles, home robots, humanoids, industrial robots, large VTOLs, professional service robots, and small drones. The model also projects aggregate robotics demand across critical components, including: cameras, lidar, radar, motors, bearings, reducers, rare earth magnets, semiconductors, edge compute, and batteries. These estimates are intended to be updated periodically as the industry develops. *For a copy of the Global Robot Model, please reach out to your Morgan Stanley representative.*

**We hope you find this unique product thought providing and insightful. And as always, we value your feedback and insights to help improve the quality and efficacy of the Almanac in the months and years to come.**

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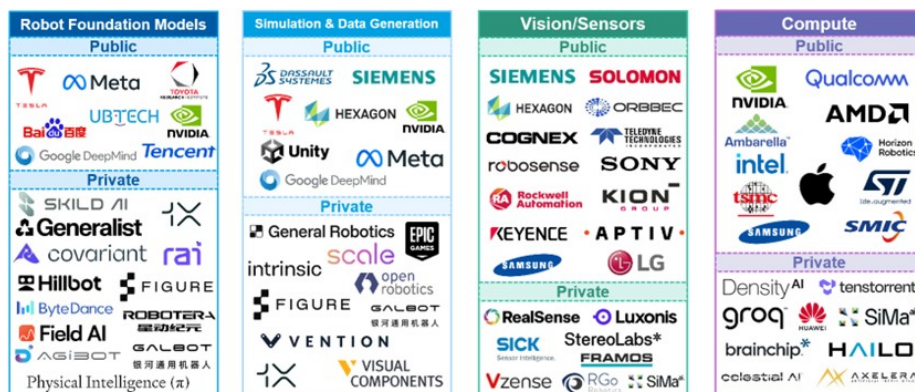
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# Public & Private Company Enablers

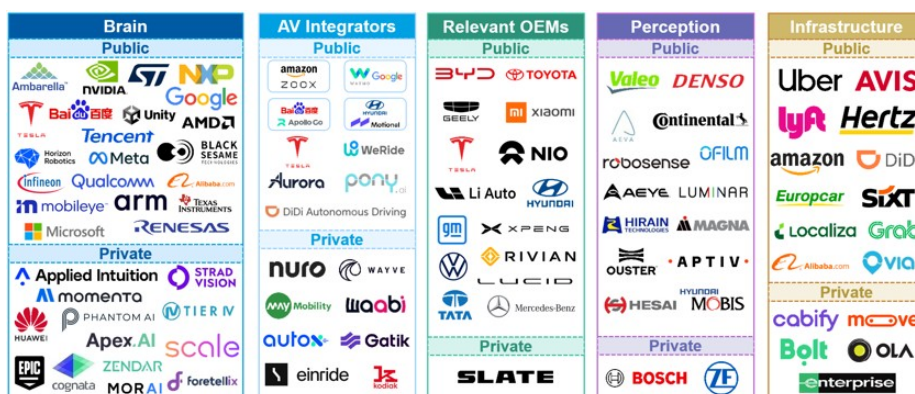
**Exhibit 1:** Global Physical AI Model Development Enablers



Note: Meant to give a representative snapshot. Does not capture 100% of exposed players.

Source: Company Websites, Morgan Stanley Research

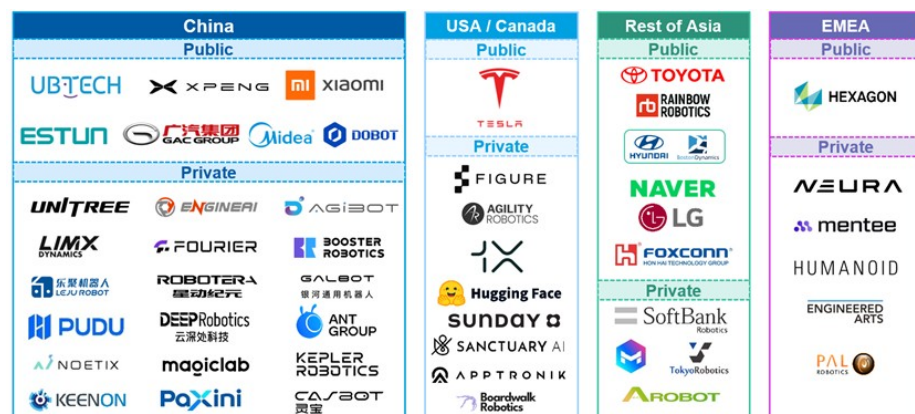
**Exhibit 2:** Global Autonomous Vehicle Enablers



Note: Meant to give a representative snapshot. Does not capture 100% of exposed players.

Source: Company Websites, Morgan Stanley Research

**Exhibit 3:** Global Humanoid Enablers (Only Includes Companies That Have Revealed Humanoids To-Date)



Note: Does not capture 100% of exposed players.. Hyundai (Korea) owns Boston Dynamics (USA). We include 1X under USA as the company recently moved its headquarters to Palo Alto.

Source: Company Websites, Morgan Stanley Research



Exhibit 4: Global Industrial Automation Enablers

| Robot OEMs (Arms, Co-Bots)                                                                                                                                                                                                                                                                                                                                                                                                               | Full Stack Automation                                                                                                                                                                                                                                                                                                                        | Software, AI, Controls                                                                                                                                                                                                                                                                                                                                                                                               | Components & Subsystems                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Public</b><br><b>ABB</b> <b>DENSO</b> <b>KAWASAKI</b><br><b>YASKAWA</b> <b>FANUC</b><br><b>EPSON</b> <b>TERADYNE</b><br><b>INOVANCE</b> <b>ESTUN</b><br><b>HYUNDAI</b> <b>DAEWOO</b> <b>DAEWOO</b><br><b>MITSUBISHI ELECTRIC</b> <b>DAEWOO</b> <b>DOOSAN</b><br><b>DELTA</b> <b>OMRON</b><br><b>Private</b><br><b>NEURA</b> <b>cobot</b><br><b>AGILE ROBOTS</b> <b>SoftBank Robotics</b><br><b>STANDARD BOTS</b> <b>Path Robotics</b> | <b>Public</b><br><b>Rockwell Automation</b><br><b>SIEMENS</b><br><b>Honeywell</b><br><b>OMRON</b> <b>INOVANCE</b><br><b>EMERSON</b> <b>Schneider Electric</b><br><b>YASKAWA</b><br><b>MITSUBISHI ELECTRIC</b> <b>ABB</b><br><b>Private</b><br><b>HADRIAN</b> <b>BECKHOFF</b><br><b>Bright Machines</b> <b>rexroth</b><br><b>MACHINA LABS</b> | <b>Public</b><br><b>NVIDIA</b> <b>DAEWOO</b><br><b>HEXAGON</b> <b>SAP</b><br><b>Palantir</b> <b>ORACLE</b><br><b>Microsoft</b> <b>amazon</b><br><b>synopsys</b> <b>IBM</b><br><b>ptc</b> <b>AUTODESK</b><br><b>Private</b><br><b>COGNITE</b> <b>TEEN TECHNOLOGIES</b><br><b>VENTION</b> <b>covariant</b><br><b>realtime</b> <b>intrinsic</b><br><b>TULIP</b> <b>LITMUS</b><br><b>BRAIN CUBE</b> <b>SIGHT MACHINE</b> | <b>Public</b><br><b>HEXAGON</b> <b>TIMKEN</b> <b>Regal Rexnord</b><br><b>HIWIN</b> <b>INOVANCE</b> <b>NSK</b><br><b>TE</b> <b>SCHAEFFLER</b> <b>H</b><br><b>Infineon</b> <b>Amphenol</b> <b>NVIDIA</b><br><b>Nabtesco</b> <b>THK</b> <b>NOTES-CLAR</b><br><b>MOONS</b> <b>绿的谐波</b> <b>Leadshine</b><br><b>Nidec</b> <b>Melexis</b> <b>Novanta</b><br><b>Sensata</b> <b>RENEASAS</b> <b>ST</b><br><b>OMNIVISION</b> <b>SKF</b> <b>TEXAS INSTRUMENTS</b> |

Note: Meant to give a representative snapshot. Does not capture 100% of exposed players. Note that ABB is spinning off its robotics business to Softbank, planned to close in mid-2026.

Source: Company Websites, Morgan Stanley Research

Exhibit 5: Global Small Drone Enablers

| Drone Manufacturing                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Control/Autonomy                                                                                                                                                                                                           | Delivery/Logistics                                                                                                                                                            | Components                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Public</b><br><b>AV</b> <b>TEAL</b> <b>Parrot</b><br><b>JOUAV</b> <b>EHANG</b> <b>Ebit Systems</b><br><b>Private</b><br><b>dji</b> <b>Skydio</b> <b>AUTEL</b><br><b>ANDURIL</b> <b>brinc</b><br><b>SKYFISH</b> <b>wingtra</b> <b>FLYABILITY</b><br><b>DELAIR</b> <b>hilio</b> <b>YUNEEC</b><br><b>Shield AI</b> <b>ASCENT</b> <b>VANTAGE ROBOTICS</b><br><b>NEROS</b> <b>VECTOR</b> <b>PERFORMANCE DRONE WORKS</b><br><b>XAG</b> <b>microdrones</b> <b>ASCENT</b> | <b>Private</b><br><b>ANDURIL</b> <b>Shield AI</b><br><b>HHelsing</b><br><b>Auterion</b><br><b>PIX4D</b><br><b>DroneDeploy</b><br><b>Mapware</b><br><b>SKYCATCH</b><br><b>RAPTOR</b><br><b>flytbase</b><br><b>propeller</b> | <b>Public</b><br><b>amazon</b><br><b>Wing</b><br><b>DHL</b><br><b>Private</b><br><b>Zipline</b><br><b>Airbound</b><br><b>FLYTREX</b><br><b>ELROY AIR</b><br><b>WINGCOPTER</b> | <b>Public</b><br><b>Nidec</b> <b>Amberella</b> <b>Qualcomm</b><br><b>TELEFONIA</b> <b>ST</b> <b>MOONS</b> <b>NVIDIA</b><br><b>FOXCONN</b> <b>KOL</b> <b>SONY</b><br><b>ZHAOWEI</b> <b>NP</b> <b>INOVANCE</b><br><b>JABIL</b> <b>UNUSUAL</b> <b>amphius</b><br><b>Private</b><br><b>2-MOTOR</b> <b>WALKSNAIL</b> <b>H3X</b><br><b>Lumenier</b> <b>BETA FPV</b> <b>RADIO MASTER</b><br><b>HOZERO</b> <b>maxon</b> <b>Shenzhen</b><br><b>MATEKSYS</b> <b>FLYCOO</b> <b>EMAX</b><br><b>Holybro</b> <b>FLIGHT</b> <b>FOKER</b> |

Note: Meant to give a representative snapshot. Does not capture 100% of exposed players.

Source: Company Websites, Morgan Stanley Research

Exhibit 6: Global Defense Tech Enablers

| Autonomous Platforms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Munitions & Effectors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Cyber & C4ISR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Supply & Manufacturing                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Public</b><br><b>AV</b> <b>TEAL</b> <b>BOEING</b> <b>NORTHROP GRUMMAN</b><br><b>GENERAL DYNAMICS</b> <b>TEXTRON</b> <b>BAE SYSTEMS</b><br><b>RHEINMETALL</b> <b>L3HARRIS</b> <b>KRATOS</b><br><b>THALES</b> <b>Ebit Systems</b> <b>LEONARDO</b><br><b>Private</b><br><b>ANDURIL</b> <b>SHIELD AI</b><br><b>Skydio</b> <b>HHelsing</b> <b>SARONIC</b><br><b>GENERAL ATOMICS</b> <b>BAYKAR</b><br><b>Swan</b> <b>NEROS</b> <b>Auterion</b><br><b>Applied Intuition</b> <b>FORTERRA</b><br><b>MILREX</b> <b>Harmattan AI</b> <b>GHOST</b><br><b>VATN</b> <b>BLUE WATER</b> <b>ARX</b><br><b>DARKHIVE</b> <b>SALDRONE</b> <b>FORTM</b> | <b>Public</b><br><b>NORTHROP GRUMMAN</b> <b>BOEING</b><br><b>LOCKHEED MARTIN</b> <b>Ebit Systems</b><br><b>RHEINMETALL</b> <b>TEXTRON</b><br><b>L3HARRIS</b> <b>BAE SYSTEMS</b><br><b>LEONARDO</b> <b>RTX</b> <b>AV</b><br><b>DRONESHIELD</b> <b>AXON</b><br><b>Private</b><br><b>ANDURIL</b> <b>EPICUS</b><br><b>HYPERSONIX</b> <b>HERMEUS</b><br><b>Castleton</b> <b>MACH INDUSTRIES</b><br><b>ACS</b> <b>ZEROMARK</b><br><b>HHelsing</b> <b>URSAMAJOR</b><br><b>ARES INDUSTRIES</b> <b>80-FEND</b><br><b>GENERAL ATOMICS</b> | <b>Public</b><br><b>Palantir</b> <b>L3HARRIS</b><br><b>BAE SYSTEMS</b> <b>LOCKHEED MARTIN</b><br><b>GENERAL DYNAMICS</b> <b>NORTHROP GRUMMAN</b><br><b>Booz   Allen   Hamilton</b> <b>Ebit Systems</b><br><b>CACI</b> <b>planet</b> <b>RTX</b><br><b>leidos</b> <b>THALES</b> <b>CYBERARK</b><br><b>Private</b><br><b>SPACEX</b> <b>SHIELD AI</b><br><b>ANDURIL</b> <b>RUNE</b><br><b>REVEAL</b> <b>HHelsing</b><br><b>VANNEVAR Labs</b> <b>Capella</b><br><b>rebellion</b> <b>CAPE</b><br><b>picogrid</b> <b>percipient.ai</b> | <b>Public</b><br><b>KRATOS</b> <b>Honeywell</b><br><b>HEICO</b> <b>TORAY</b><br><b>Trimble</b> <b>Palantir</b><br><b>Regal Rexnord</b> <b>HEXCEL</b> <b>Nidec</b><br><b>SOLVAY</b> <b>DAEWOO</b><br><b>TITOMIC</b> <b>REDWIRE</b><br><b>IPERIONX</b> <b>Hanwha</b><br><b>Private</b><br><b>Solugen</b> <b>HADRIAN</b><br><b>KoBold Metals</b> <b>VulcanForms</b><br><b>ASTRO MECHANICA</b> <b>gecko</b> <b>FIREHAWK</b><br><b>CHAOS</b> <b>DIVERGENT3D</b><br><b>Alloyed</b> <b>MACHINA LABS</b> |

Note: Meant to give a representative snapshot. Does not capture 100% of exposed players. C4ISR = Command, Control, Communications, Computers, Intelligence, Surveillance, Reconnaissance. "Munitions & Effectors" include counter-drone systems.

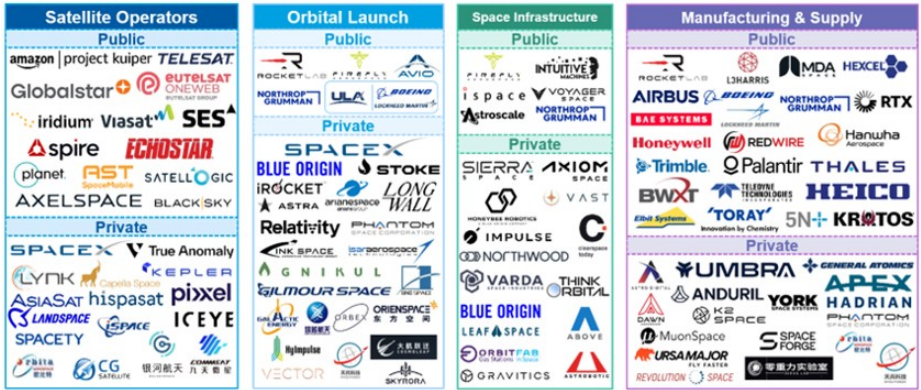
Source: Company Websites, Morgan Stanley Research

Exhibit 7: Global Next-Gen Air Transport Enablers



Note: Meant to give a representative snapshot. Does not capture 100% of exposed players. Point to Point Space largely includes players building spaceplanes or reusable rockets.  
Source: Company Websites, Morgan Stanley Research

Exhibit 8: Global Space Industry Enablers



Note: Meant to give a representative snapshot. Does not capture 100% of exposed players. Does not include government agencies or government-owned enterprises. Infrastructure includes space stations, rovers, landers, ground stations, etc.  
Source: Company Websites, Morgan Stanley Research

# MS Global Robot Model (GRoM) Details

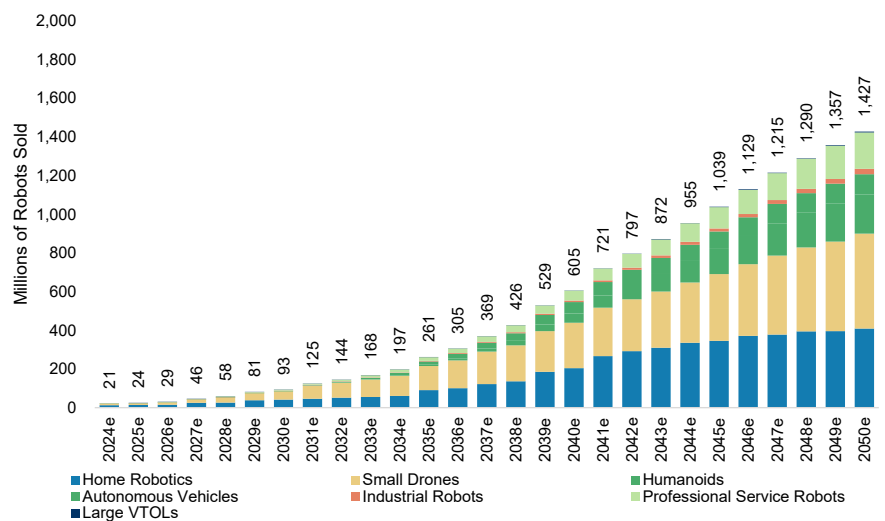
For a copy of the Global Robot Model, please reach out to your Morgan Stanley representative.

## Sub-Models included within Global Robot Model:

- US, China, and Global Humanoid Model
- China and Global Professional Service Robot Model
- China and Global Industrial Robot Model
- US, China, and Global Large VTOL Model
- US, China, and Global Small Drone Model
- US and Global Home Robot Model
- Global Autonomous Vehicle Model
- Global Robot Content Model (Cameras, LiDAR, Radar, Motors, Bearings, Reducers, Analog Semis, Edge Compute, Rare Earth Magnets, Batteries)

## Global Robotics Industry Estimates

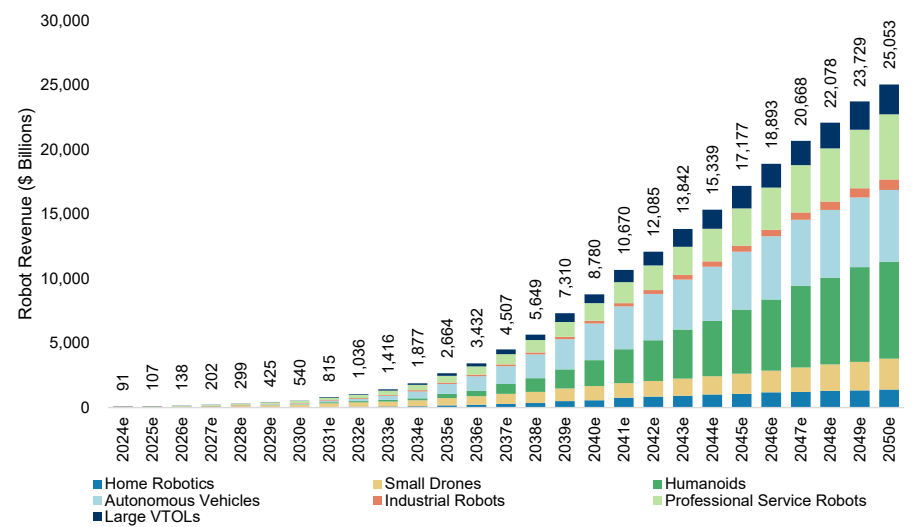
**Exhibit 9:** We See 1.4 Billion in Annual Robot Unit Sales by 2050



Note: Final Hardware Sales Only- Software/Services, Maintenance, Supply Chains, etc. could theoretically multiply this figure.

Source: Morgan Stanley Research estimates

**Exhibit 10:** We See \$25 Trillion in Global Robot Revenues by 2050 (Final Hardware Sales Only)



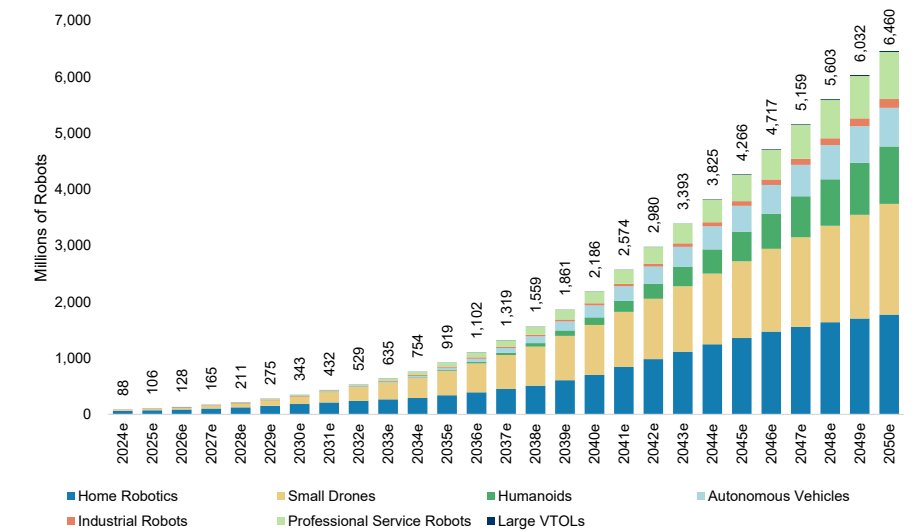
Note: Final Hardware Sales Only- Software/Services, Maintenance, Supply Chains, etc. could theoretically multiply this figure.  
Source: Morgan Stanley Research estimates

**Exhibit 11:** Bull/Bear/Base Global Revenue and Hardware Sales Estimates

|                       |  |      |       |       |       |       |       |       |       |         |         |         |         |          | Flex 2035-2050 |          |  |  |  |
|-----------------------|--|------|-------|-------|-------|-------|-------|-------|-------|---------|---------|---------|---------|----------|----------------|----------|--|--|--|
| Bull/Bear Analysis    |  | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2031e | 2032e   | 2033e   | 2034e   | 2035e   | 2040e    | 2045e          | 2050e    |  |  |  |
| Total Global Robotics |  |      |       |       |       |       |       |       |       |         |         |         |         |          |                |          |  |  |  |
| Unit Sales (mns)      |  |      |       |       |       |       |       |       |       |         |         |         |         |          |                |          |  |  |  |
| Bull                  |  | 21   | 24    | 29    | 46    | 58    | 81    | 93    | 125   | 144     | 168     | 197     | 261     | 940      | 1,795          | 2,345    |  |  |  |
| Base                  |  | 21   | 24    | 29    | 46    | 58    | 81    | 93    | 125   | 144     | 168     | 197     | 261     | 905      | 1,039          | 1,427    |  |  |  |
| Bear                  |  | 21   | 24    | 29    | 46    | 58    | 81    | 93    | 125   | 144     | 168     | 197     | 261     | 413      | 595            | 717      |  |  |  |
| Revenue (\$ bns)      |  |      |       |       |       |       |       |       |       |         |         |         |         |          |                |          |  |  |  |
| Bull                  |  | \$91 | \$107 | \$138 | \$202 | \$299 | \$425 | \$540 | \$815 | \$1,036 | \$1,416 | \$1,877 | \$2,664 | \$14,696 | \$28,076       | \$40,311 |  |  |  |
| Base                  |  | \$91 | \$107 | \$138 | \$202 | \$299 | \$425 | \$540 | \$815 | \$1,036 | \$1,416 | \$1,877 | \$2,664 | \$8,780  | \$17,177       | \$25,063 |  |  |  |
| Bear                  |  | \$91 | \$107 | \$138 | \$202 | \$299 | \$425 | \$540 | \$815 | \$1,036 | \$1,416 | \$1,877 | \$2,664 | \$5,234  | \$10,201       | \$12,102 |  |  |  |

Source: Morgan Stanley Research estimates

**Exhibit 12:** We See 6.5 Billion in Robots in Operation Globally by 2050



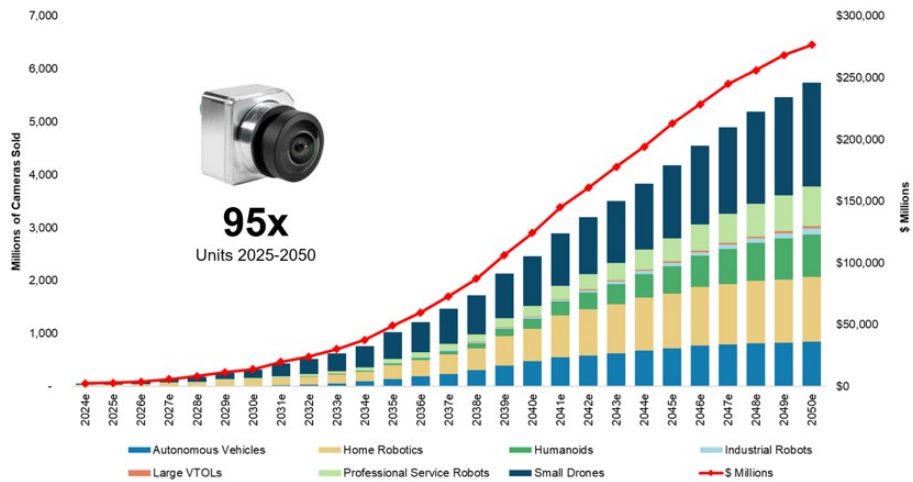
Note: Final Hardware Sales Only- Software/Services, Maintenance, Supply Chains, etc. could theoretically multiply this figure.  
Source: Morgan Stanley Research estimates





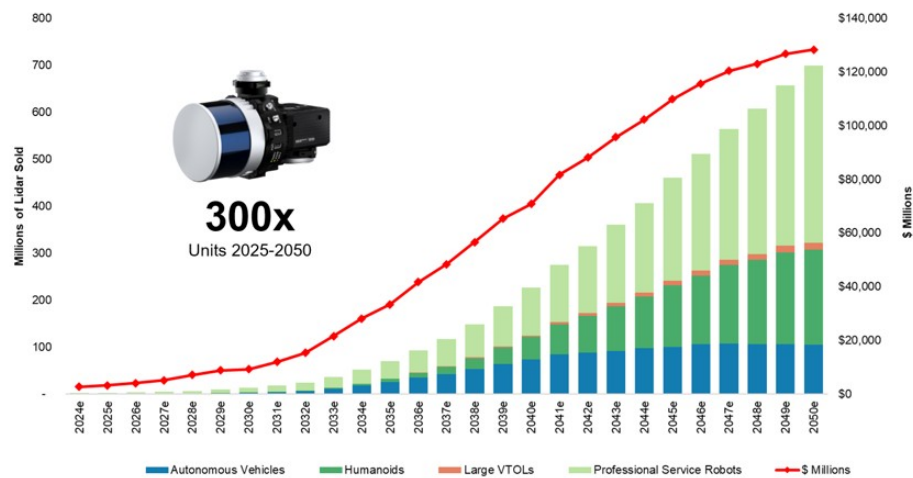
# Robotics Content Estimates

Exhibit 16: Component Demand: Cameras



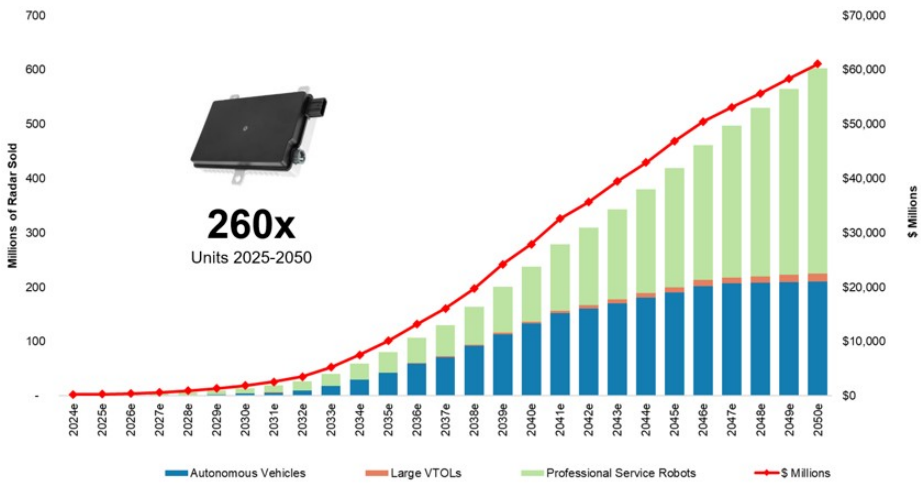
Source: Morgan Stanley Research estimates

Exhibit 17: Component Demand: LiDAR



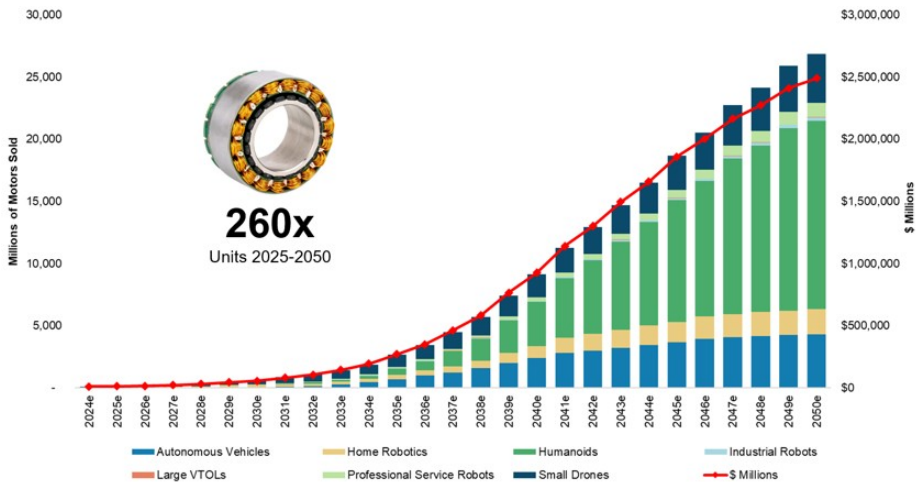
Source: Morgan Stanley Research estimates

Exhibit 18: Component Demand: Radar



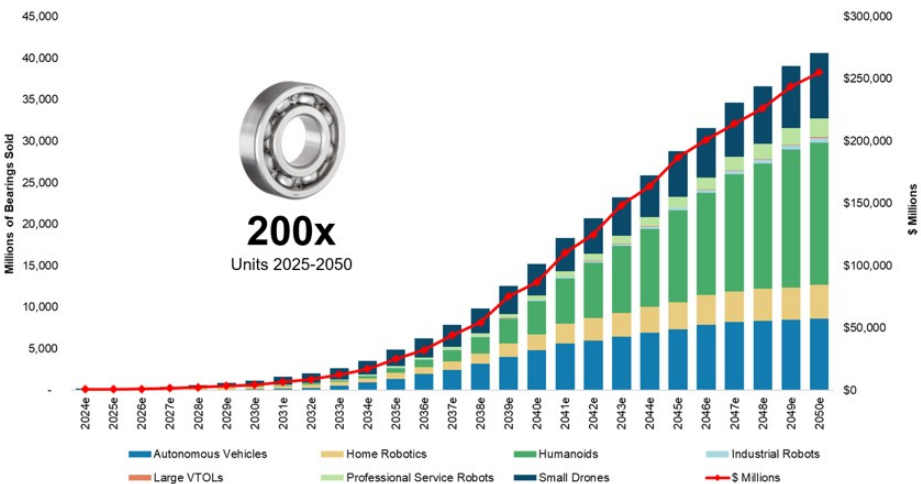
Source: Morgan Stanley Research estimates

Exhibit 19: Component Demand: Electric Motors



Source: Morgan Stanley Research estimates

Exhibit 20: Component Demand: Bearings



Source: Morgan Stanley Research estimates

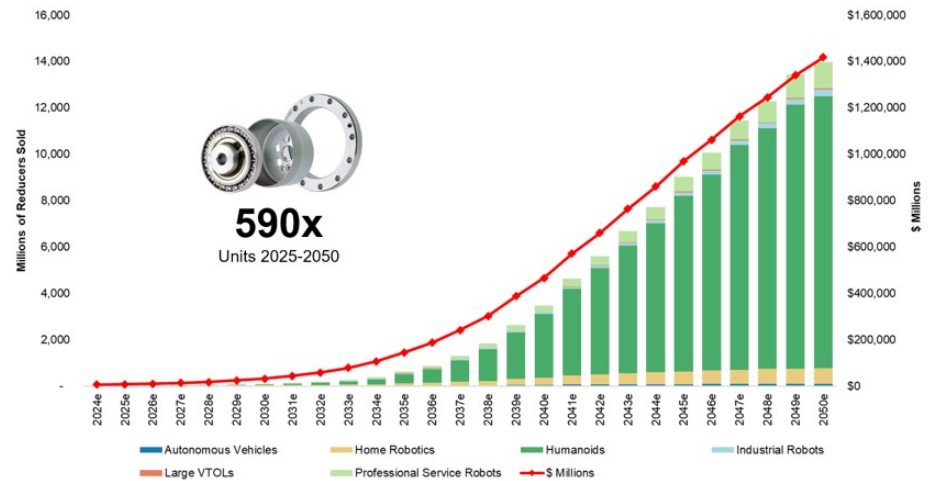
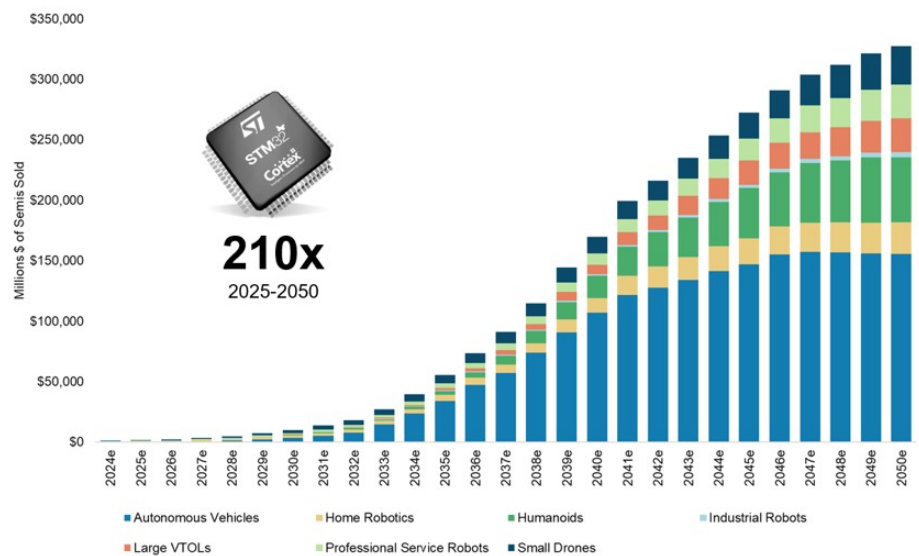
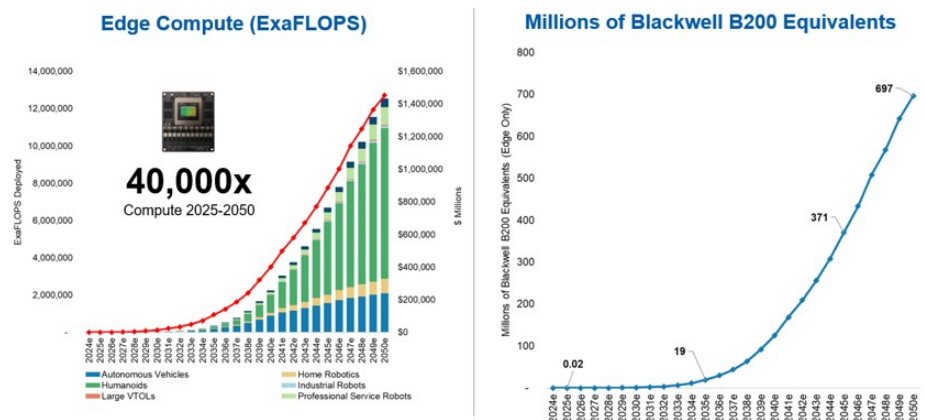
**Exhibit 21: Component Demand: Reducers****Exhibit 22: Component Demand: Analog Semis & MCU's****Exhibit 23: Component Demand: Edge Compute**



Exhibit 24: Component Demand: Rare Earth Magnets (NdFeB, SmCo, etc.)

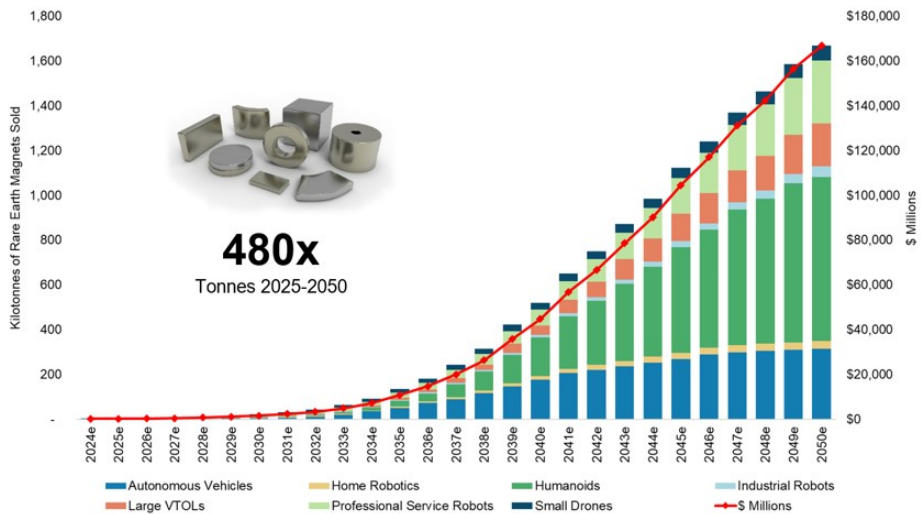
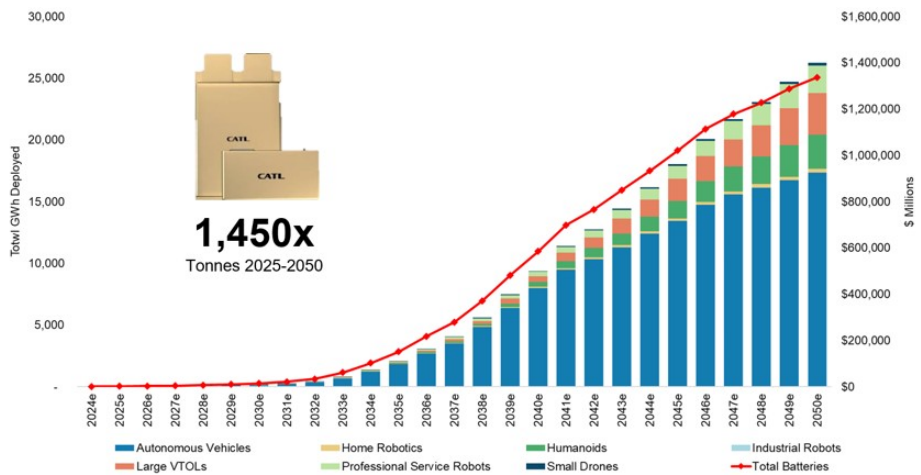


Exhibit 25: Component Demand: Batteries (All Chemistries)



# Industrial Robots

## Exhibit 26: Snapshot of Global Industrial Robot Estimates

| Robots By Region (k's)                   | 2017         | 2018         | 2019         | 2020         | 2021         | 2022         | 2023         | 2024         | 2025e        | 2026e        | 2027e        | 2028e        | 2029e        | 2030e        | 2035e         | 2040e         | 2050e          |
|------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|----------------|
| China                                    | 501          | 650          | 789          | 906          | 1,226        | 1,501        | 1,755        | 2,027        | 2,346        | 2,710        | 3,126        | 3,599        | 4,135        | 4,742        | 9,132         | 16,751        | 48,466         |
| Japan                                    | 297          | 318          | 355          | 374          | 393          | 414          | 435          | 451          | 466          | 482          | 508          | 546          | 608          | 689          | 1,463         | 3,239         | 10,216         |
| Europe                                   | 498          | 543          | 582          | 614          | 675          | 729          | 778          | 830          | 879          | 938          | 1,031        | 1,133        | 1,268        | 1,444        | 3,538         | 9,408         | 41,781         |
| United States                            | 262          | 285          | 300          | 314          | 342          | 365          | 382          | 394          | 408          | 426          | 463          | 507          | 570          | 653          | 1,578         | 4,002         | 14,720         |
| RoW                                      | 987          | 845          | 711          | 709          | 843          | 895          | 932          | 952          | 965          | 1,028        | 1,083        | 1,152        | 1,236        | 1,351        | 2,539         | 5,905         | 30,857         |
| <b>Installed Base</b>                    | <b>2,153</b> | <b>2,441</b> | <b>2,737</b> | <b>3,027</b> | <b>3,479</b> | <b>3,904</b> | <b>4,282</b> | <b>4,664</b> | <b>5,083</b> | <b>5,565</b> | <b>6,211</b> | <b>6,936</b> | <b>7,817</b> | <b>8,879</b> | <b>18,220</b> | <b>39,311</b> | <b>108,549</b> |
| <b>Total Robots Sold/Installed</b>       | <b>400</b>   | <b>423</b>   | <b>387</b>   | <b>390</b>   | <b>526</b>   | <b>553</b>   | <b>541</b>   | <b>542</b>   | <b>628</b>   | <b>695</b>   | <b>878</b>   | <b>1,005</b> | <b>1,227</b> | <b>1,452</b> | <b>3,419</b>  | <b>7,835</b>  | <b>28,093</b>  |
| Replacement                              | 113          | 107          | 91           | 100          | 74           | 128          | 164          | 160          | 198          | 204          | 251          | 280          | 347          | 391          | 861           | 2,188         | 11,040         |
| Replacement Rate                         | 6%           | 5%           | 4%           | 4%           | 2%           | 4%           | 4%           | 4%           | 4%           | 4%           | 5%           | 5%           | 5%           | 5%           | 6%            | 7%            | 8%             |
| Implied Average Life (Check)             | 16           | 20           | 27           | 27           | 41           | 27           | 24           | 27           | 24           | 25           | 22           | 22           | 22           | 20           | 20            | 18            | 13             |
| Y/Y Growth                               | 16%          | 15%          | 12%          | 11%          | 15%          | 12%          | 10%          | 9%           | 9%           | 10%          | 11%          | 12%          | 13%          | 14%          | 16%           | 17%           | 12%            |
| <b>Net Adds By Region (k's)</b>          | <b>2017</b>  | <b>2018</b>  | <b>2019</b>  | <b>2020</b>  | <b>2021</b>  | <b>2022</b>  | <b>2023</b>  | <b>2024</b>  | <b>2025e</b> | <b>2026e</b> | <b>2027e</b> | <b>2028e</b> | <b>2029e</b> | <b>2030e</b> | <b>2035e</b>  | <b>2040e</b>  | <b>2050e</b>   |
| China                                    | 152          | 149          | 139          | 167          | 270          | 275          | 254          | 272          | 319          | 364          | 416          | 473          | 536          | 607          | 1,090         | 1,857         | 4,463          |
| Japan                                    | 10           | 21           | 37           | 19           | 19           | 21           | 21           | 15           | 16           | 16           | 26           | 38           | 62           | 81           | 215           | 476           | 869            |
| Europe                                   | 38           | 45           | 39           | 32           | 61           | 54           | 49           | 53           | 48           | 59           | 93           | 102          | 135          | 176          | 629           | 1,684         | 3,703          |
| United States                            | 12           | 23           | 15           | 14           | 28           | 23           | 17           | 12           | 14           | 19           | 36           | 44           | 64           | 83           | 268           | 679           | 1,122          |
| RoW                                      | 75           | 78           | 66           | 58           | 74           | 62           | 37           | 31           | 32           | 33           | 55           | 68           | 84           | 115          | 355           | 950           | 6,596          |
| <b>Total Net Additions to Base</b>       | <b>287</b>   | <b>316</b>   | <b>296</b>   | <b>290</b>   | <b>452</b>   | <b>426</b>   | <b>378</b>   | <b>382</b>   | <b>429</b>   | <b>492</b>   | <b>626</b>   | <b>725</b>   | <b>881</b>   | <b>1,062</b> | <b>2,558</b>  | <b>5,647</b>  | <b>17,653</b>  |
| Check                                    | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK            | OK            | OK             |
| <b>Annual Installs By Industry (k's)</b> | <b>2017</b>  | <b>2018</b>  | <b>2019</b>  | <b>2020</b>  | <b>2021</b>  | <b>2022</b>  | <b>2023</b>  | <b>2024</b>  | <b>2025e</b> | <b>2026e</b> | <b>2027e</b> | <b>2028e</b> | <b>2029e</b> | <b>2030e</b> | <b>2035e</b>  | <b>2040e</b>  | <b>2050e</b>   |
| Automotive                               | 123          | 120          | 102          | 14           | 117          | 130          | 135          | 128          | 146          | 162          | 204          | 224          | 266          | 338          | 795           | 1,623         | 6,535          |
| % of Total                               | 29.8%        | 26.4%        | 21.5%        | 22.3%        | 24.6%        | 25.0%        | 23.3%        | 23.3%        | 23.3%        | 23.3%        | 23.3%        | 23.3%        | 23.3%        | 23.3%        | 23.3%         | 23.3%         | 23.3%          |
| Electronics                              | 122          | 105          | 89           | 110          | 143          | 157          | 126          | 129          | 149          | 185          | 209          | 239          | 292          | 345          | 813           | 1,863         | 6,681          |
| % of Total                               | 24.8%        | 23.0%        | 28.0%        | 27.1%        | 28.4%        | 23.2%        | 23.8%        | 23.8%        | 23.8%        | 23.8%        | 23.8%        | 23.8%        | 23.8%        | 23.8%        | 23.8%         | 23.8%         | 23.8%          |
| Metal & Machinery                        | 44           | 44           | 52           | 44           | 68           | 66           | 77           | 79           | 91           | 101          | 128          | 146          | 178          | 211          | 497           | 1,138         | 4,081          |
| % of Total                               | 10.4%        | 12.4%        | 11.3%        | 13.0%        | 12.0%        | 14.2%        | 14.5%        | 14.5%        | 14.5%        | 14.5%        | 14.5%        | 14.5%        | 14.5%        | 14.5%        | 14.5%         | 14.5%         | 14.5%          |
| Plastics & Chemicals                     | 21           | 20           | 18           | 19           | 65           | 24           | 22           | 23           | 27           | 29           | 37           | 43           | 52           | 62           | 145           | 332           | 1,190          |
| % of Total                               | 4.7%         | 4.7%         | 4.9%         | 4.8%         | 4.3%         | 4.1%         | 4.2%         | 4.2%         | 4.2%         | 4.2%         | 4.2%         | 4.2%         | 4.2%         | 4.2%         | 4.2%          | 4.2%          | 4.2%           |
| Food & Beverage                          | 9            | 12           | 11           | 12           | 15           | 15           | 15           | 15           | 17           | 19           | 24           | 28           | 34           | 40           | 95            | 219           | 760            |
| % of Total                               | 2.8%         | 2.8%         | 3.1%         | 2.9%         | 2.7%         | 2.7%         | 2.8%         | 2.8%         | 2.8%         | 2.8%         | 2.8%         | 2.8%         | 2.8%         | 2.8%         | 2.8%          | 2.8%          | 2.8%           |
| Other                                    | 81           | 116          | 115          | 121          | 157          | 165          | 166          | 170          | 197          | 218          | 276          | 316          | 386          | 456          | 1,074         | 2,461         | 8,825          |
| % of Total                               | 27.4%        | 29.7%        | 31.0%        | 29.9%        | 28.1%        | 30.7%        | 31.4%        | 31.4%        | 31.4%        | 31.4%        | 31.4%        | 31.4%        | 31.4%        | 31.4%        | 31.4%         | 31.4%         | 31.4%          |
| <b>Total Installations</b>               | <b>400</b>   | <b>423</b>   | <b>387</b>   | <b>390</b>   | <b>526</b>   | <b>553</b>   | <b>541</b>   | <b>542</b>   | <b>628</b>   | <b>695</b>   | <b>878</b>   | <b>1,005</b> | <b>1,227</b> | <b>1,452</b> | <b>3,419</b>  | <b>7,835</b>  | <b>28,093</b>  |
| Check                                    | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK           | OK            | OK            | OK             |

Source: Morgan Stanley Research estimates

## Professional Service Robots

**Exhibit 27:** Snapshot of Global Professional Service Robot Estimates

| Sales By Application (k's)     | 2023       | 2024         | 2025e        | 2026e        | 2027e        | 2028e        | 2029e         | 2030e         | 2035e         | 2040e          | 2050e          |
|--------------------------------|------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------|----------------|----------------|
| Transportation & Logistics     | 336        | 423          | 550          | 726          | 981          | 1,324        | 1,787         | 2,413         | 8,948         | 20,795         | 52,887         |
| % of Total                     | 48%        | 48%          | 48%          | 48%          | 49%          | 49%          | 50%           | 50%           | 48%           | 41%            | 28%            |
| Y/Y Growth                     |            | 26%          | 30%          | 32%          | 35%          | 35%          | 35%           | 35%           | 25%           | 16%            | 8%             |
| Hospitality                    | 137        | 173          | 225          | 297          | 400          | 541          | 730           | 985           | 3,653         | 8,492          | 21,597         |
| % of Total                     | 19%        | 19%          | 20%          | 20%          | 20%          | 20%          | 20%           | 20%           | 20%           | 17%            | 11%            |
| Y/Y Growth                     |            | 26%          | 30%          | 32%          | 35%          | 35%          | 35%           | 35%           | 25%           | 16%            | 8%             |
| Agriculture                    | 64         | 80           | 108          | 146          | 197          | 266          | 359           | 485           | 2,176         | 5,855          | 15,021         |
| % of Total                     | 9%         | 9%           | 9%           | 10%          | 10%          | 10%          | 10%           | 10%           | 12%           | 11%            | 8%             |
| Y/Y Growth                     |            | 26%          | 35%          | 35%          | 35%          | 35%          | 35%           | 35%           | 35%           | 15%            | 8%             |
| Professional Cleaning          | 83         | 105          | 126          | 151          | 181          | 218          | 261           | 314           | 780           | 1,909          | 5,492          |
| % of Total                     | 12%        | 12%          | 11%          | 10%          | 9%           | 8%           | 7%            | 6%            | 4%            | 4%             | 3%             |
| Y/Y Growth                     |            | 26%          | 20%          | 20%          | 20%          | 20%          | 20%           | 20%           | 20%           | 18%            | 8%             |
| Medical & Healthcare           | 55         | 69           | 89           | 118          | 159          | 215          | 290           | 392           | 1,756         | 7,872          | 42,373         |
| % of Total                     | 8%         | 8%           | 8%           | 8%           | 8%           | 8%           | 8%            | 8%            | 9%            | 15%            | 22%            |
| Y/Y Growth                     |            | 26%          | 30%          | 32%          | 35%          | 35%          | 35%           | 35%           | 35%           | 35%            | 10%            |
| Other & New Applications       | 30         | 37           | 50           | 68           | 91           | 128          | 179           | 251           | 1,349         | 6,048          | 50,995         |
| % of Total                     | 4%         | 4%           | 4%           | 4%           | 5%           | 5%           | 5%            | 5%            | 7%            | 12%            | 27%            |
| Y/Y Growth                     |            | 26%          | 35%          | 35%          | 35%          | 40%          | 40%           | 40%           | 40%           | 35%            | 15%            |
| <b>Total Robots Sold (k's)</b> | <b>705</b> | <b>887</b>   | <b>1,149</b> | <b>1,506</b> | <b>2,010</b> | <b>2,691</b> | <b>3,607</b>  | <b>4,839</b>  | <b>18,659</b> | <b>50,970</b>  | <b>188,365</b> |
| Y/Y Growth                     |            | 26%          | 29%          | 31%          | 33%          | 34%          | 34%           | 34%           | 28%           | 21%            | 10%            |
| Sales By Region (k's)          | 2023       | 2024         | 2025e        | 2026e        | 2027e        | 2028e        | 2029e         | 2030e         | 2035e         | 2040e          | 2050e          |
| China                          | 352        | 444          | 629          | 881          | 1,224        | 1,601        | 2,088         | 2,710         | 8,250         | 19,842         | 65,442         |
| US                             | 70         | 89           | 115          | 151          | 201          | 296          | 397           | 532           | 2,426         | 7,646          | 28,255         |
| Europe                         | 106        | 133          | 172          | 226          | 302          | 431          | 577           | 774           | 3,359         | 10,194         | 37,673         |
| RoW                            | 176        | 222          | 293          | 348          | 484          | 664          | 893           | 1,203         | 4,624         | 13,289         | 56,995         |
| <b>Total Robots Sold</b>       | <b>705</b> | <b>887</b>   | <b>1,149</b> | <b>1,506</b> | <b>2,010</b> | <b>2,691</b> | <b>3,607</b>  | <b>4,839</b>  | <b>18,659</b> | <b>50,970</b>  | <b>188,365</b> |
| China                          | 50%        | 50%          | 55%          | 59%          | 61%          | 59%          | 58%           | 56%           | 44%           | 39%            | 35%            |
| US                             | 10%        | 10%          | 10%          | 10%          | 10%          | 11%          | 11%           | 11%           | 13%           | 15%            | 15%            |
| Europe                         | 15%        | 15%          | 15%          | 15%          | 15%          | 16%          | 16%           | 16%           | 18%           | 20%            | 20%            |
| RoW                            | 25%        | 25%          | 20%          | 16%          | 14%          | 14%          | 15%           | 17%           | 25%           | 26%            | 30%            |
| % of Total                     | 100%       | 100%         | 100%         | 100%         | 100%         | 100%         | 100%          | 100%          | 100%          | 100%           | 100%           |
| Check                          | OK         | OK           | OK           | OK           | OK           | OK           | OK            | OK            | OK            | OK             | OK             |
| Installed Base (k's)           | 2023       | 2024         | 2025e        | 2026e        | 2027e        | 2028e        | 2029e         | 2030e         | 2035e         | 2040e          | 2050e          |
| China                          | 580        | 970          | 1,570        | 2,404        | 3,556        | 5,050        | 6,986         | 9,487         | 33,917        | 88,891         | 309,438        |
| % of Total                     | 62%        | 55%          | 55%          | 56%          | 58%          | 58%          | 58%           | 58%           | 49%           | 42%            | 37%            |
| US                             | 108        | 211          | 371          | 641          | 1,109        | 1,733        | 2,402         | 3,298         | 13,782        | 41,752         | 167,583        |
| % of Total                     | 12%        | 12%          | 13%          | 15%          | 18%          | 20%          | 20%           | 20%           | 20%           | 20%            | 20%            |
| Europe                         | 108        | 211          | 371          | 641          | 1,109        | 1,733        | 2,402         | 3,298         | 13,782        | 41,752         | 167,583        |
| % of Total                     | 12%        | 12%          | 13%          | 15%          | 18%          | 20%          | 20%           | 20%           | 20%           | 20%            | 20%            |
| RoW                            | 124        | 368          | 544          | 590          | 386          | 149          | 221           | 408           | 7,428         | 36,565         | 193,311        |
| % of Total                     | 14%        | 21%          | 19%          | 14%          | 6%           | 2%           | 2%            | 2%            | 11%           | 18%            | 23%            |
| <b>Installed Base (k's)</b>    | <b>900</b> | <b>1,760</b> | <b>2,856</b> | <b>4,276</b> | <b>6,159</b> | <b>8,665</b> | <b>12,012</b> | <b>16,491</b> | <b>68,908</b> | <b>208,761</b> | <b>837,916</b> |
| Total Robots Sold              | 705        | 887          | 1,149        | 1,506        | 2,010        | 2,691        | 3,607         | 4,839         | 18,659        | 50,970         | 188,365        |
| Total Replacement              |            | 27           | 53           | 86           | 128          | 185          | 260           | 360           | 2,645         | 13,721         | 105,741        |
| Replacement Rate               |            | 3%           | 3%           | 3%           | 3%           | 3%           | 3%            | 3%            | 5%            | 8%             | 14%            |
| Implied Avg Life (Check)       |            | 33.3         | 33.3         | 33.3         | 33.3         | 33.3         | 33.3          | 33.3          | 20.0          | 12.5           | 7.1            |
| Check                          | OK         | OK           | OK           | OK           | OK           | OK           | OK            | OK            | OK            | OK             | OK             |

Source: Morgan Stanley Research estimates

## Large VTOLs

Exhibit 28: Snapshot of Global Large VTOL Estimates

| 2020 VTOLs / Population |      |                     |                     | VTOLs / Population |        |        |        |        |        |        |        |        |        |        |
|-------------------------|------|---------------------|---------------------|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Country                 | Cont | Region              | Income Group        | 2020               | 2021   | 2022   | 2023   | 2024   | 2025   | 2026   | 2027   | 2028   | 2029   | 2030   |
| USA                     | NA   | North America       | High Income         | 8.23%              | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |
| China                   | AS   | East Asia & Pacific | Upper-Middle Income | 0.000%             | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |
| Low Income              |      |                     |                     | 0.000%             | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |
| Lower-Middle Income     |      |                     |                     | 0.000%             | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |
| Upper-Middle Income     |      |                     |                     | 0.000%             | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |
| High Income             |      |                     |                     | 0.000%             | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |
| Global Total            |      |                     |                     | 8.17%              | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% | 0.000% |

| Country Info (World Bank Classifications) |      |                     |                     | VTOL Adoptability |            |      |      |       |       |       |        |        |        |         |
|-------------------------------------------|------|---------------------|---------------------|-------------------|------------|------|------|-------|-------|-------|--------|--------|--------|---------|
| Country Name                              | Cont | Region              | Income Group        | 2020 Population   | 2020 VTOLs | 2021 | 2022 | 2023  | 2024  | 2025  | 2026   | 2027   | 2028   | 2029    |
| United States of America                  | USA  | North America       | High Income         | 385,412,816       | 2,491,838  | -    | 5    | 15    | 48    | 139   | 448    | 1,130  | 2,318  | 39,346  |
| China                                     | CN   | East Asia & Pacific | Upper-Middle Income | 1,251,432,312     | 2,821,408  | 243  | 745  | 1,473 | 2,838 | 5,082 | 8,853  | 14,314 | 22,833 | 165,948 |
| Low Income                                |      |                     |                     | 1,222,838,989     | 400,404    | -    | -    | -     | -     | -     | -      | -      | 288    | 5,553   |
| Lower-Middle Income                       |      |                     |                     | 4,186,898,548     | 3,411,318  | -    | -    | -     | -     | 539   | 1,339  | 2,827  | 55,572 | 437,551 |
| Upper-Middle Income (excl. China)         |      |                     |                     | 1,580,367,014     | 3,440,721  | -    | -    | -     | -     | 190   | 613    | 1,515  | 3,180  | 60,793  |
| High Income (excl. USA)                   |      |                     |                     | 1,028,899,987     | 4,492,985  | -    | -    | 92    | 277   | 888   | 2,183  | 4,554  | 84,583 | 631,179 |
| Global Total                              |      |                     |                     | 13,444,838,216    | 16,880,083 | 243  | 750  | 1,485 | 2,977 | 5,888 | 11,141 | 22,494 | 38,777 | 416,378 |

| Country Info (World Bank Classifications) |      |                     |                     | VTOLs Solidification |      |       |       |       |       |        |         |         |           |           |
|-------------------------------------------|------|---------------------|---------------------|----------------------|------|-------|-------|-------|-------|--------|---------|---------|-----------|-----------|
| Country Name                              | Cont | Region              | Income Group        | 2023                 | 2024 | 2025  | 2026  | 2027  | 2028  | 2029   | 2030    | 2031    | 2032      | 2033      |
| United States of America                  | USA  | North America       | High Income         | 5                    | 15   | 31    | 84    | 312   | 671   | 1,267  | 17,866  | 132,562 | 530,762   | 1,115,318 |
| China                                     | CN   | East Asia & Pacific | Upper-Middle Income | 508                  | 688  | 1,483 | 2,378 | 3,804 | 6,005 | 8,485  | 68,172  | 175,200 | 771,546   | 1,115,318 |
| Low Income                                |      |                     |                     | -                    | -    | -     | -     | -     | -     | -      | 288     | 2,821   | (8,859)   | 89,870    |
| Lower-Middle Income                       |      |                     |                     | -                    | -    | -     | -     | 839   | 818   | 1,554  | 27,004  | 154,811 | 739,521   | 1,115,318 |
| Upper-Middle Income (excl. China)         |      |                     |                     | -                    | -    | -     | -     | 190   | 427   | 820    | 1,741   | 30,357  | 163,025   | 731,983   |
| High Income (excl. USA)                   |      |                     |                     | -                    | -    | 92    | 187   | 617   | 1,321 | 2,480  | 41,991  | 218,962 | 943,896   | 1,115,318 |
| Global Total                              |      |                     |                     | 513                  | 803  | 1,487 | 2,848 | 8,008 | 9,789 | 18,954 | 188,432 | 935,811 | 1,615,318 | 1,115,318 |

Source: Morgan Stanley Research estimates

Small Drones

Exhibit 29: Snapshot of Global Small Drones Estimates

See USA Tab

Inputs for Non-USA/China

| 2050 Drones / Population |       |       |       | Drones / Population |       |       |       |       |       |       |       |       |  |  |
|--------------------------|-------|-------|-------|---------------------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                          | 2023  | 2024e | 2025e | 2026e               | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |       |  |  |
| USA                      | 39.2% | 0.5%  | 0.3%  | 0.3%                | 0.4%  | 0.7%  | 1.1%  | 1.3%  | 2.1%  | 2.1%  | 13.7% | 20.2% |  |  |
| China                    | 42.0% | 0.2%  | 0.1%  | 0.2%                | 1.1%  | 1.4%  | 1.7%  | 2.2%  | 2.7%  | 8.2%  | 19.3% | 42.2% |  |  |
| Low Income               | 2.0%  | 0.2%  | 0.2%  | 0.2%                | 0.3%  | 0.1%  | 0.1%  | 0.2%  | 0.2%  | 0.7%  | 1.2%  | 2.0%  |  |  |
| Lower Middle Income      | 15.1% | 0.1%  | 0.1%  | 0.1%                | 0.2%  | 0.2%  | 0.2%  | 0.3%  | 1.1%  | 0.6%  | 8.1%  | 11.1% |  |  |
| Upper Middle Income      | 22.7% | 0.1%  | 0.2%  | 0.2%                | 0.3%  | 0.2%  | 0.2%  | 1.2%  | 1.6%  | 0.4%  | 10.3% | 22.7% |  |  |
| High Income              | 30.2% | 0.1%  | 0.2%  | 0.2%                | 0.4%  | 0.2%  | 1.1%  | 1.6%  | 8.1%  | 2.2%  | 13.7% | 30.2% |  |  |
| Global Total             | 20.4% | 0.5%  | 0.2%  | 0.3%                | 0.4%  | 0.6%  | 0.8%  | 1.2%  | 1.6%  | 4.9%  | 9.7%  | 20.4% |  |  |

| Country Info (World Bank Classifications) |      |                     |                     | Drone Adoptability |             |        |        |        |        |        |        |        |         |         |         |           |
|-------------------------------------------|------|---------------------|---------------------|--------------------|-------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|-----------|
| Country Name                              | Code | Region              | Income Group        | 2050 Population    | 2050 Drones | 2023   | 2024e  | 2025e  | 2026e  | 2027e  | 2028e  | 2029e  | 2030e   | 2035e   | 2040e   | 2050e     |
| United States of America                  | USA  | North America       | High income         | 380,413            | 114,944     | 399    | 702    | 1,001  | 1,405  | 2,204  | 3,721  | 5,800  | 7,586   | 25,630  | 60,629  | 114,944   |
| China                                     | CHN  | East Asia & Pacific | Upper middle income | 1,265,452          | 530,970     | 7,219  | 9,599  | 12,307 | 15,582 | 19,606 | 24,509 | 30,521 | 37,563  | 112,802 | 259,174 | 530,970   |
| Low Income                                |      |                     |                     | 1,222,839          | 24,463      | 72     | 142    | 207    | 307    | 485    | 819    | 1,268  | 1,789   | 6,576   | 13,919  | 24,463    |
| Lower Middle Income                       |      |                     |                     | 4,166,606          | 629,484     | 1,671  | 3,276  | 4,708  | 6,895  | 10,955 | 17,894 | 27,121 | 37,003  | 131,625 | 262,256 | 629,484   |
| Upper Middle Income (excl. China)         |      |                     |                     | 1,580,667          | 358,071     | 1,087  | 2,143  | 3,064  | 4,461  | 7,049  | 11,452 | 17,260 | 23,417  | 80,994  | 157,670 | 358,071   |
| High Income (excl. USA)                   |      |                     |                     | 1,038,669          | 316,877     | 1,091  | 2,127  | 3,020  | 4,365  | 6,864  | 11,666 | 16,478 | 22,366  | 75,109  | 142,472 | 316,877   |
| Installed Home Drone Base                 |      |                     |                     | 9,844,936          | 1,968,786   | 11,269 | 17,888 | 24,367 | 35,065 | 47,283 | 65,461 | 86,339 | 130,054 | 432,735 | 887,116 | 1,968,786 |

| Country Info (World Bank Classifications) |      |                     |                     | Drones Sold/Installed (k%) |       |        |        |        |        |        |         |         |         |       |
|-------------------------------------------|------|---------------------|---------------------|----------------------------|-------|--------|--------|--------|--------|--------|---------|---------|---------|-------|
| Country Name                              | Code | Region              | Income Group        | 2023                       | 2024e | 2025e  | 2026e  | 2027e  | 2028e  | 2029e  | 2030e   | 2035e   | 2040e   | 2050e |
| United States of America                  | USA  | North America       | High income         | 362                        | 377   | 664    | 1,000  | 1,679  | 2,288  | 2,668  | 6,977   | 13,274  | 30,148  |       |
| China                                     | CHN  | East Asia & Pacific | Upper middle income | 2,729                      | 3,469 | 4,430  | 5,607  | 6,965  | 8,656  | 10,706 | 33,654  | 68,124  | 122,665 |       |
| Low Income                                |      |                     |                     | 78                         | 81    | 123    | 221    | 379    | 529    | 632    | 1,960   | 3,820   | 2,349   |       |
| Lower Middle Income                       |      |                     |                     | 1,789                      | 1,793 | 2,705  | 4,818  | 8,144  | 11,195 | 13,136 | 38,131  | 70,606  | 164,467 |       |
| Upper Middle Income (excl. China)         |      |                     |                     | 1,167                      | 1,157 | 1,734  | 3,078  | 5,178  | 7,068  | 8,228  | 23,143  | 41,675  | 92,199  |       |
| High Income (excl. USA)                   |      |                     |                     | 1,196                      | 1,127 | 1,677  | 2,989  | 4,967  | 6,729  | 7,987  | 21,163  | 37,143  | 79,262  |       |
| Global Total                              |      |                     |                     | 7,311                      | 7,831 | 11,243 | 17,434 | 27,916 | 38,466 | 43,118 | 131,928 | 245,454 | 481,508 |       |

Source: Morgan Stanley Research estimates



Home Robots

Exhibit 30: Snapshot of Global Home Robot Estimates

See USA Tab  
Inputs for Non-USA/China

| 2020 Robots/Household |  |  |  | Robots per Household |       |       |       |       |       |       |       |       |       |
|-----------------------|--|--|--|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                       |  |  |  | 2023                 | 2024e | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e |
| USA                   |  |  |  | 0.07                 | 0.08  | 0.08  | 0.09  | 0.10  | 0.12  | 0.14  | 0.16  | 0.20  | 0.25  |
| China                 |  |  |  | 0.03                 | 0.04  | 0.04  | 0.05  | 0.06  | 0.07  | 0.09  | 0.11  | 0.15  | 0.20  |
| Low Income            |  |  |  | 0.00                 | 0.00  | 0.00  | 0.00  | 0.00  | 0.01  | 0.01  | 0.02  | 0.03  | 0.05  |
| Lower Middle Income   |  |  |  | 0.01                 | 0.01  | 0.01  | 0.01  | 0.01  | 0.02  | 0.03  | 0.04  | 0.07  | 0.10  |
| Upper Middle Income   |  |  |  | 0.02                 | 0.02  | 0.03  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  | 0.12  | 0.18  |
| High Income           |  |  |  | 0.04                 | 0.05  | 0.07  | 0.08  | 0.09  | 0.10  | 0.11  | 0.12  | 0.20  | 0.25  |
| Global Total          |  |  |  | 0.08                 | 0.09  | 0.09  | 0.09  | 0.09  | 0.09  | 0.09  | 0.09  | 0.14  | 0.20  |

| Country Info (World Bank Classifications) |      |        |              | Robot Acceptability (%) |             |        |        |         |         |         |         |         |         |
|-------------------------------------------|------|--------|--------------|-------------------------|-------------|--------|--------|---------|---------|---------|---------|---------|---------|
| Country Name                              | Code | Region | Income Group | 2020 Households         | 2019 Robots | 2023   | 2024e  | 2025e   | 2026e   | 2027e   | 2028e   | 2029e   | 2030e   |
| United States of America                  |      |        |              | 145,844                 | 9,895       | 10,432 | 11,218 | 12,182  | 13,510  | 15,746  | 16,822  | 21,218  | 34,988  |
| China                                     |      |        |              | 451,947                 | 474,545     | 15,115 | 17,761 | 20,253  | 22,732  | 30,244  | 35,200  | 45,138  | 55,011  |
| Low Income                                |      |        |              | 235,029                 | 14,713      | 17     | 34     | 69      | 142     | 284     | 569     | 1,065   | 1,218   |
| Lower Middle Income                       |      |        |              | 308,309                 | 295,561     | 1,417  | 2,147  | 3,261   | 4,953   | 7,518   | 11,408  | 19,232  | 27,228  |
| Upper Middle Income (excl. China)         |      |        |              | 427,534                 | 356,648     | 5,743  | 7,701  | 9,893   | 11,707  | 15,107  | 19,751  | 23,837  | 27,862  |
| High Income (excl. USA)                   |      |        |              | 411,774                 | 452,351     | 21,035 | 25,321 | 29,541  | 33,729  | 37,663  | 42,170  | 46,377  | 50,574  |
| Installed Home Robot Base                 |      |        |              | 2,851,438               | 1,773,467   | 63,916 | 83,416 | 104,036 | 124,636 | 165,446 | 195,144 | 254,571 | 383,211 |

| Country Info (World Bank Classifications) |      |        |              | Robots Sold/Installed (%) |        |        |        |        |        |        |        |         |         |
|-------------------------------------------|------|--------|--------------|---------------------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|
| Country Name                              | Code | Region | Income Group | 2023                      | 2024e  | 2025e  | 2026e  | 2027e  | 2028e  | 2029e  | 2030e  | 2035e   | 2040e   |
| United States of America                  |      |        |              | 1,149                     | 1,289  | 1,581  | 1,998  | 3,114  | 4,278  | 3,904  | 9,286  | 21,106  | 40,779  |
| China                                     |      |        |              | 3,251                     | 3,380  | 3,093  | 8,762  | 6,922  | 12,402 | 13,710 | 25,493 | 53,665  | 101,061 |
| Low Income                                |      |        |              | 18                        | 37     | 77     | 449    | 323    | 287    | 243    | 614    | 1,505   | 3,429   |
| Lower Middle Income                       |      |        |              | 787                       | 1,221  | 1,872  | 2,837  | 4,379  | 8,822  | 9,631  | 11,420 | 23,098  | 75,500  |
| Upper Middle Income (excl. China)         |      |        |              | 2,188                     | 2,377  | 2,147  | 4,643  | 5,063  | 5,489  | 6,152  | 14,074 | 38,432  | 94,790  |
| High Income (excl. USA)                   |      |        |              | 5,128                     | 5,486  | 5,813  | 6,079  | 6,684  | 7,199  | 8,159  | 29,664 | 68,858  | 93,213  |
| Installations / Sales                     |      |        |              | 12,325                    | 13,736 | 16,493 | 24,149 | 26,487 | 35,187 | 41,719 | 85,652 | 204,664 | 405,748 |

Source: Morgan Stanley Research estimates

Humanoids

Exhibit 31: Snapshot of Global Humanoid Estimates

| Market Size Analysis                     |       |       |       |       |       |       |       |       |       |        |         |         |         |         |         |         |
|------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|---------|---------|---------|---------|---------|---------|
| USA                                      | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | -     | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.7    | 1.8     | 3.3     | 5.0     | 6.5     | 7.4     | 7.4     |
| Replacement Units (mn)                   | -     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | 0.0   | 0.0   | 0.0    | 0.2     | 0.2     | 0.7     | 1.8     | 3.5     | 6.7     |
| Total Humanoid Units Sold (mn)           | -     | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.7    | 1.8     | 3.5     | 5.7     | 8.3     | 10.9    | 13.4    |
| Average Selling Price Per Humanoid (\$k) | 200.0 | 187.5 | 175.0 | 162.5 | 150.0 | 137.5 | 125.0 | 122.3 | 115.1 | 106.3  | 101.9   | 95.9    | 90.2    | 84.9    | 81.5    | 78.3    |
| USA Revenue (\$bn)                       | -     | 0.0   | 0.0   | 1.3   | 2.9   | 6.3   | 4.2   | 21.3  | 25.3  | 74.7   | 182.8   | 334.7   | 517.6   | 703.3   | 888.6   | 1,048.6 |
| China                                    | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.2   | 0.4   | 1.3   | 3.9    | 10.4    | 22.0    | 31.4    | 31.9    | 24.9    | 15.3    |
| Replacement Units (mn)                   | -     | N/A   | N/A   | N/A   | N/A   | N/A   | 0.0   | 0.0   | 0.1   | 0.2    | 0.4     | 1.4     | 4.1     | 10.8    | 23.3    | 36.5    |
| Total Humanoid Units Sold (mn)           | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.2   | 0.4   | 1.4   | 4.1    | 10.8    | 23.3    | 35.5    | 42.7    | 48.2    | 50.8    |
| Average Selling Price Per Humanoid (\$k) | 40.0  | 46.5  | 42.5  | 39.6  | 36.9  | 34.3  | 32.0  | 27.7  | 24.1  | 22.1   | 21.8    | 20.9    | 20.5    | 20.1    | 20.5    | 20.9    |
| China Revenue (\$bn)                     | 0.1   | 0.3   | 0.6   | 1.1   | 2.1   | 3.5   | 6.1   | 12.8  | 32.9  | 89.6   | 251.3   | 487.4   | 728.3   | 855.6   | 966.8   | 1,059.6 |
| Low Income Countries                     | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | -     | -     | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.1    | 0.2     | 0.3     | 0.6     | 1.0     | 1.5     | 1.8     |
| Replacement Units (mn)                   | -     | -     | -     | -     | -     | -     | N/A   | N/A   | 0.0   | 0.0    | 0.0     | 0.0     | 0.1     | 0.2     | 0.3     | 0.7     |
| Total Humanoid Units Sold (mn)           | -     | -     | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.1    | 0.2     | 0.3     | 0.7     | 1.2     | 1.8     | 2.5     |
| Average Selling Price Per Humanoid (\$k) | 49.0  | 45.6  | 42.5  | 39.6  | 36.9  | 34.3  | 32.0  | 27.3  | 24.1  | 22.1   | 21.8    | 20.9    | 20.5    | 20.1    | 20.5    | 21.3    |
| LI Country Revenue (\$bn)                | -     | -     | -     | -     | -     | -     | 0.1   | 1.4   | 1.8   | 4.8    | 10.0    | 32.7    | 59.2    | 106.0   | 204.7   | 353.3   |
| Lower Middle Income Countries            | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 1.4    | 2.6     | 4.5     | 8.8     | 14.5    | 21.9    | 26.4    |
| Replacement Units (mn)                   | -     | -     | -     | -     | N/A   | N/A   | N/A   | 0.0   | 0.0   | 0.2    | 0.6     | 1.5     | 2.8     | 5.1     | 10.2    | 17.3    |
| Total Humanoid Units Sold (mn)           | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 1.5    | 2.8     | 5.1     | 10.2    | 17.3    | 27.0    | 43.0    |
| Average Selling Price Per Humanoid (\$k) | 40.0  | 45.6  | 42.5  | 39.6  | 36.9  | 34.3  | 32.0  | 27.7  | 24.1  | 22.1   | 21.8    | 20.9    | 20.5    | 20.1    | 20.5    | 21.3    |
| LMI Country Revenue (\$bn)               | -     | -     | -     | -     | 0.7   | 1.2   | 1.8   | 8.1   | 14.8  | 32.8   | 59.9    | 105.9   | 204.6   | 348.7   | 551.5   | 764.9   |
| Upper Middle Income Countries (Ex-China) | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 1.4    | 2.6     | 4.5     | 8.8     | 14.6    | 21.8    | 24.6    |
| Replacement Units (mn)                   | -     | -     | -     | -     | N/A   | N/A   | N/A   | 0.0   | 0.0   | 0.2    | 0.6     | 1.5     | 2.8     | 5.1     | 10.3    | 17.4    |
| Total Humanoid Units Sold (mn)           | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 1.5    | 2.8     | 5.1     | 10.3    | 17.4    | 26.9    | 36.4    |
| Average Selling Price Per Humanoid (\$k) | 40.0  | 45.6  | 42.5  | 39.6  | 36.9  | 34.3  | 32.0  | 27.7  | 24.1  | 22.1   | 21.8    | 20.9    | 20.5    | 20.1    | 20.5    | 21.3    |
| UMI Country Revenue (\$bn)               | -     | -     | -     | -     | 0.7   | 1.2   | 1.8   | 8.1   | 14.8  | 32.8   | 59.9    | 105.9   | 204.6   | 348.7   | 549.7   | 760.6   |
| High Income Countries (Ex-USA)           | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | -     | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.5   | 0.6   | 1.9    | 4.6     | 9.5     | 14.4    | 16.5    | 20.5    | 22.3    |
| Replacement Units (mn)                   | -     | N/A   | N/A   | N/A   | N/A   | N/A   | 0.0   | 0.0   | 0.1   | 0.5    | 0.6     | 2.0     | 5.1     | 10.1    | 16.4    | 23.6    |
| Total Humanoid Units Sold (mn)           | -     | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.5   | 0.6   | 2.0    | 5.1     | 10.1    | 16.4    | 23.6    | 30.8    | 44.0    |
| Average Selling Price Per Humanoid (\$k) | 200.0 | 187.5 | 175.0 | 162.5 | 150.0 | 137.5 | 125.0 | 122.3 | 115.1 | 106.3  | 101.9   | 95.9    | 90.2    | 84.9    | 81.5    | 78.3    |
| HIC Country Revenue (\$bn)               | -     | 1.8   | 2.6   | 4.9   | 8.7   | 14.3  | 14.3  | 58.5  | 70.1  | 214.9  | 520.6   | 967.9   | 1,476.7 | 2,004.2 | 2,491.0 | 3,043.0 |
| Global                                   | 2024  | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2032e | 2034e | 2036e  | 2038e   | 2040e   | 2042e   | 2044e   | 2046e   | 2050e   |
| New Humanoid Units Adopted (mn)          | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.3   | 0.4   | 1.4   | 3.3   | 9.4    | 22.0    | 43.9    | 69.0    | 87.0    | 97.9    | 99.9    |
| Replacement Units (mn)                   | -     | -     | -     | -     | 0.0   | 0.0   | 0.0   | 0.2   | 0.4   | 1.5    | 3.5     | 8.0     | 23.5    | 47.4    | 78.8    | 119.5   |
| Total Humanoid Units Sold (mn)           | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.3   | 0.4   | 1.6   | 3.5   | 8.8    | 23.5    | 47.4    | 78.8    | 110.6   | 145.4   | 178.7   |
| Average Selling Price Per Humanoid (\$k) | 40.0  | 138.3 | 119.8 | 105.7 | 70.5  | 80.0  | 64.6  | 70.2  | 48.8  | 46.0   | 42.4    | 40.0    | 36.8    | 37.9    | 37.7    | 37.7    |
| Global Revenue (\$bn)                    | 0.1   | 2.4   | 4.0   | 6.4   | 12.1  | 25.7  | 27.7  | 101.0 | 158.0 | 446.0  | 1,058.1 | 2,009.1 | 3,150.3 | 4,293.3 | 5,505.2 | 6,728.9 |
| Global Humanoid Stock (k's)              | 2     | 22    | 55    | 115   | 287   | 597   | 1,015 | 1,332 | 8,793 | 24,436 | 61,424  | 137,923 | 284,400 | 430,447 | 624,103 | 826,858 |

Source: Morgan Stanley Research estimates

Autonomous Vehicles

Exhibit 32: Snapshot of Global AV Estimates

| Global Autonomous Vehicle Model (Millions) |      |       |       |       |       |       |       |       |       |       |
|--------------------------------------------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| USA                                        | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.2   | 0.3   | 0.5   | 1.7   | 5.3   | 1.9   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.9   | 2.6   | 16.7  |
| L4 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.2   | 0.5   | 1.0   | 5.7   | 20.1  | 39.6  |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.8   | 9.6   | 73.2  |
| Europe                                     | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 3.8   | 2.3   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 2.2   | 20.3  |
| L4 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 8.5   | 77.7  |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 3.4   | 93.3  |
| China                                      | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.1   | 0.2   | 0.5   | 0.7   | 2.4   | 6.0   | 2.4   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.2   | 3.6   | 21.5  |
| L4 Park                                    | 0.0  | 0.0   | 0.1   | 0.1   | 0.4   | 0.8   | 1.5   | 8.5   | 26.6  | 60.8  |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 2.5   | 14.1  | 104.4 |
| India                                      | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.4   | 1.5   | 5.5   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.5   | 5.8   |
| L4 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.8   | 4.7   | 25.9  |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.1   | 19.9  |
| Japan                                      | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.5   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   |
| L4 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 3.9   |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.5   | 0.8   |
| Korea                                      | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.6   | 1.3   | 0.1   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 1.6   |
| L4 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.3   | 5.9   | 6.4   |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.4   | 7.7   |
| ROW                                        | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.2   | 5.9   | 17.4  | 8.0   |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 2.7   | 18.6  |
| L4 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.4   | 11.7  | 64.1  | 93.3  |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 3.9   | 81.7  |
| Global                                     | 2024 | 2025e | 2026e | 2027e | 2028e | 2029e | 2030e | 2035e | 2040e | 2050e |
| New L4 AV's                                | 0.0  | 0.0   | 0.1   | 0.1   | 0.5   | 1.0   | 1.5   | 14.8  | 47.1  | 20.5  |
| New L5 AV's                                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 2.1   | 12.0  | 84.6  |
| L4 Park                                    | 0.0  | 0.0   | 0.1   | 0.2   | 0.7   | 1.6   | 3.0   | 36.6  | 175.7 | 307.7 |
| L5 Park                                    | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 4.4   | 33.0  | 380.9 |
| New L4/L5 AV's                             | 0.0  | 0.0   | 0.1   | 0.1   | 0.5   | 1.0   | 1.5   | 16.9  | 59.1  | 105.2 |
| L4/L5 Park                                 | 0.0  | 0.0   | 0.1   | 0.2   | 0.7   | 1.6   | 3.0   | 41.0  | 208.7 | 688.6 |

Source: Morgan Stanley Research estimates

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(as of November 30, 2025)

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| Stock Rating Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1501              | 41%        | 389                              | 46%            | 26%                  | 661                                               | 41%                   |
| Equal-weight/Hold     | 1608              | 44%        | 379                              | 44%            | 24%                  | 736                                               | 46%                   |
| Not-Rated/Hold        | 4                 | 0%         | 1                                | 0%             | 25%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 567               | 15%        | 83                               | 10%            | 15%                  | 212                                               | 13%                   |
| Total                 | 3,680             |            | 852                              |                |                      | 1610                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Ariana Salvatore.

## INDUSTRY COVERAGE: Autos & Shared Mobility

| COMPANY (TICKER)                                   | RATING (AS OF)  | PRICE* (12/05/2025) |
|----------------------------------------------------|-----------------|---------------------|
| <b>Adam Jonas, CFA</b>                             |                 |                     |
| Adient PLC (ADNT.N)                                | U (03/17/2021)  | \$18.52             |
| American Axle & Manufacturing Holdings Inc (AXL.N) | O (02/28/2022)  | \$6.31              |
| Aptiv Plc (APTV.N)                                 | U (02/06/2024)  | \$76.37             |
| Avis Budget Group Inc (CAR.O)                      | O (06/20/2023)  | \$134.85            |
| BorgWarner Inc. (BWA.N)                            | O (05/15/2023)  | \$43.30             |
| Ferrari NV (RACE.N)                                | O (05/09/2019)  | \$393.21            |
| Ford Motor Company (F.N)                           | E (09/25/2024)  | \$13.03             |
| General Motors Company (GM.N)                      | E (12/10/2024)  | \$76.05             |
| Hertz Global Holdings Inc (HTZ.O)                  | E (02/08/2024)  | \$5.06              |
| Lear Corporation (LEA.N)                           | O (05/10/2024)  | \$107.78            |
| Lucid Group Inc (LCID.O)                           | E (03/17/2025)  | \$13.42             |
| Magna International Inc. (MGA.N)                   | E (09/25/2024)  | \$49.03             |
| Mobileye Global Inc (MBLY.O)                       | E (08/02/2024)  | \$11.70             |
| PHINIA INC (PHIN.N)                                | E (09/25/2024)  | \$54.41             |
| Quantumscape Corp (QS.N)                           | NR (11/21/2024) | \$12.55             |
| Rivian Automotive, Inc. (RIVN.O)                   | E (09/25/2024)  | \$17.95             |
| Tesla Inc (TSLA.O)                                 | O (09/10/2023)  | \$455.00            |
| Visteon Corporation (VC.O)                         | E (06/01/2022)  | \$99.48             |
| <b>Daniela M Haigian</b>                           |                 |                     |
| Asbury Automotive Group Inc (ABG.N)                | E (09/25/2024)  | \$238.90            |
| AutoNation Inc. (AN.N)                             | O (09/25/2024)  | \$215.92            |
| Carmax Inc (KMX.N)                                 | E (11/10/2025)  | \$38.82             |
| Carvana Co (CVNA.N)                                | O (08/07/2025)  | \$399.77            |

|                                        |                |          |
|----------------------------------------|----------------|----------|
| Group 1 Automotive, Inc (GPI.N)        | O (09/25/2024) | \$404.21 |
| Lithia Motors Inc. (LAD.N)             | E (09/25/2024) | \$328.39 |
| Penske Automotive Group, Inc (PAG.N)   | O (09/25/2024) | \$164.93 |
| Sonic Automotive Inc (SAH.N)           | E (09/25/2024) | \$64.37  |
| <b>Javier Martinez de Olcoz Cerdan</b> |                |          |
| Goodyear Tire & Rubber Company (GT.O)  | U (09/28/2025) | \$8.58   |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Aerospace

| COMPANY (TICKER)                        | RATING (AS OF) | PRICE* (12/05/2025) |
|-----------------------------------------|----------------|---------------------|
| <b>Kristine T Liwag</b>                 |                |                     |
| AerCap Holdings NV (AER.N)              | E (06/07/2024) | \$139.99            |
| Beta Technologies Inc (BETA.N)          | O (12/01/2025) | \$28.39             |
| Boeing Co. (BA.N)                       | E (01/10/2023) | \$201.89            |
| CAE Inc. (CAE.TO)                       | E (08/16/2024) | C\$37.82            |
| Curtiss-Wright Corp. (CW.N)             | O (08/06/2023) | \$544.60            |
| FTAI Aviation Ltd (FTAI.O)              | O (01/11/2024) | \$177.02            |
| Heico Corp (HEI.N)                      | E (06/04/2024) | \$312.20            |
| Hexcel Corp (HXL.N)                     | U (02/22/2024) | \$76.60             |
| Howmet (HWM.N)                          | O (12/03/2021) | \$190.98            |
| Joby Aviation Inc (JOBY.N)              | E (04/11/2025) | \$15.26             |
| Loar Holdings Inc (LOAR.N)              | O (05/23/2025) | \$67.25             |
| Moog Inc. (MOGa.N)                      | E (11/22/2023) | \$237.47            |
| RBC Bearings Inc. (RBC.N)               | O (12/03/2021) | \$443.44            |
| Rocket Lab USA Inc (RKL.B.O)            | E (07/15/2024) | \$49.06             |
| RTX Corp (RTX.N)                        | O (04/23/2025) | \$171.10            |
| Spirit AeroSystems Holdings Inc (SPR.N) | ++             | \$39.50             |
| StandardAero Inc. (SARO.N)              | E (10/28/2024) | \$26.56             |
| Teledyne Technologies Inc. (TDY.N)      | E (12/12/2022) | \$522.30            |
| TransDigm Group Inc. (TDG.N)            | O (09/10/2021) | \$1,346.86          |
| Virgin Galactic Holdings Inc (SPCE.N)   | U (11/22/2023) | \$4.31              |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: China Internet and Other Services

| COMPANY (TICKER)                    | RATING (AS OF) | PRICE* (12/05/2025) |
|-------------------------------------|----------------|---------------------|
| <b>Eddy Wang, CFA</b>               |                |                     |
| Autohome Inc (ATHM.N)               | E (02/09/2023) | \$23.88             |
| Full Truck Alliance Co. Ltd (YMM.N) | O (07/05/2023) | \$11.52             |
| JD.com, Inc. (JD.O)                 | U (11/10/2025) | \$29.90             |
| Kanzhun Ltd (BZ.O)                  | O (08/04/2021) | \$21.77             |
| KE Holdings Inc (BEKE.N)            | O (03/16/2022) | \$17.18             |
| PDD Holdings Inc (PDD.O)            | O (03/02/2023) | \$117.62            |
| Vipshop Holdings Ltd (VIPS.N)       | E (02/24/2022) | \$20.12             |
| <b>Gary Yu</b>                      |                |                     |
| Alibaba Group Holding (BABA.N)      | O (02/24/2025) | \$158.32            |
| Baidu Inc (BIDU.O)                  | E (05/17/2024) | \$125.66            |
| Meituan (3690.HK)                   | O (08/29/2024) | HK\$99.05           |
| Tencent Holdings Ltd. (0700.HK)     | O (03/19/2020) | HK\$610.00          |
| <b>Rebecca Xu</b>                   |                |                     |
| Hello Group Inc (MOMO.O)            | U (05/16/2024) | \$7.06              |

|                                     |                |           |
|-------------------------------------|----------------|-----------|
| HUYA Inc (HUYA.N)                   | E (05/16/2024) | \$2.97    |
| IQIYI Inc (IQ.O)                    | E (01/19/2023) | \$2.09    |
| JOYY Inc. (JOYY.O)                  | E (06/02/2022) | \$63.86   |
| Weibo Corp (WB.O)                   | U (05/17/2024) | \$10.42   |
| <b>Yang Liu</b>                     |                |           |
| Bilibili Inc (BILI.O)               | E (01/07/2022) | \$25.80   |
| Kuaishou Technology (1024.HK)       | E (07/04/2024) | HK\$69.20 |
| NetEase, Inc (NTES.O)               | O (01/08/2025) | \$139.78  |
| Tongcheng Travel Holdings (0780.HK) | O (01/04/2019) | HK\$22.42 |
| Trip.com Group Ltd (TCOM.O)         | O (05/17/2021) | \$71.75   |

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\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: China Industrials

| COMPANY (TICKER)                                   | RATING (AS OF) | PRICE* (12/05/2025) |
|----------------------------------------------------|----------------|---------------------|
| <b>Chelsea Wang</b>                                |                |                     |
| China Railway Group (601390.SS)                    | E (05/12/2022) | Rmb5.52             |
| China Railway Group (0390.HK)                      | E (08/11/2025) | HK\$3.89            |
| China State Construction Engineering (601668.SS)   | U (08/11/2025) | Rmb5.33             |
| Han's Laser (002008.SZ)                            | O (10/02/2025) | Rmb37.92            |
| Hefei Meyer Optoelectronic Technology (002690.SZ)  | E (09/08/2025) | Rmb20.61            |
| iRay Technology Company Limited (688301.SS)        | E (01/16/2025) | Rmb101.35           |
| Neway Valve (Suzhou) Co., Ltd (603699.SS)          | O (09/12/2025) | Rmb48.46            |
| Shanghai BOCHU Electronic Technology (688188.SS)   | O (08/22/2024) | Rmb132.62           |
| Shenzhen Envicool Technology Co Ltd (002837.SZ)    | O (08/19/2024) | Rmb71.61            |
| <b>Sheng Zhong</b>                                 |                |                     |
| Beijing Geekplus Technology Co., Ltd. (2590.HK)    | O (08/07/2025) | HK\$24.80           |
| Centre Testing International Group (300012.SZ)     | E (11/18/2024) | Rmb13.77            |
| CRRC Corp Ltd (1766.HK)                            | E (08/25/2025) | HK\$6.10            |
| CRRC Corp Ltd (601766.SS)                          | E (06/30/2022) | Rmb7.11             |
| DR Laser (300776.SZ)                               | E (12/17/2021) | Rmb59.66            |
| Estun Automation Co Ltd (002747.SZ)                | U (06/30/2022) | Rmb22.55            |
| Haitian International Holdings Limited (1882.HK)   | E (09/08/2025) | HK\$22.74           |
| Hongfa Technology Co Ltd (600885.SS)               | O (05/23/2023) | Rmb29.14            |
| Jiangsu Guomao Reducer Co Ltd (603915.SS)          | U (01/08/2025) | Rmb17.26            |
| Jiangsu Hengli Hydraulic Co.Ltd (601100.SS)        | O (05/23/2023) | Rmb112.98           |
| Jingsheng Mechanical & Electrical Co (300316.SZ)   | U (01/08/2025) | Rmb35.35            |
| Leader Harmonious Drive Systems (688017.SS)        | E (05/19/2025) | Rmb157.49           |
| Sany Heavy Industry Co., Ltd. (600031.SS)          | O (01/08/2025) | Rmb21.30            |
| Shenzhen Inovance Technology (300124.SZ)           | O (05/19/2025) | Rmb73.16            |
| Shenzhen SC New Energy Technology Corp (300724.SZ) | U (09/08/2025) | Rmb84.24            |
| Sinotruk (Hong Kong) Limited (3808.HK)             | E (05/19/2025) | HK\$30.34           |
| Suzhou Maxwell Technologies Co Ltd (300751.SZ)     | U (09/15/2023) | Rmb125.02           |
| Times Electric (3898.HK)                           | O (08/30/2022) | HK\$38.80           |
| WeiChai Power (2338.HK)                            | E (09/08/2025) | HK\$19.98           |
| WeiChai Power (000338.SZ)                          | E (09/08/2025) | Rmb17.70            |
| Wuxi Autowell Technology Co Ltd (688516.SS)        | U (09/08/2025) | Rmb39.55            |
| Wuxi Lead Intelligent (300450.SZ)                  | O (09/08/2025) | Rmb48.34            |
| Zhejiang Dingli Machinery Co Ltd. (603338.SS)      | O (11/05/2025) | Rmb60.65            |
| Zhejiang Hangke Technology (688006.SS)             | O (09/08/2025) | Rmb28.05            |
| Zhejiang Shuanghuan Driveline Co. Ltd. (002472.SZ) | O (08/25/2023) | Rmb41.10            |
| Zoomlion Heavy Industry (1157.HK)                  | O (09/08/2025) | HK\$7.90            |
| Zoomlion Heavy Industry (000157.SZ)                | O (09/08/2025) | Rmb8.64             |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Machinery & Construction

| COMPANY (TICKER)                                 | RATING (AS OF) | PRICE* (12/05/2025) |
|--------------------------------------------------|----------------|---------------------|
| <b>Angel Castillo</b>                            |                |                     |
| AGCO Corp (AGCO.N)                               | U (04/16/2025) | \$105.26            |
| Allison Transmission Holdings Inc (ALSN.N)       | E (01/08/2024) | \$92.54             |
| Caterpillar Inc (CAT.N)                          | U (08/07/2025) | \$603.17            |
| CNH Industrial NV (CNH.N)                        | O (12/17/2024) | \$9.44              |
| CRH (CRH.N)                                      | O (07/09/2024) | \$119.49            |
| Cummins Inc (CMI.N)                              | O (01/08/2024) | \$510.65            |
| Deere & Co. (DE.N)                               | O (01/08/2024) | \$475.11            |
| Donaldson Company Inc. (DCI.N)                   | E (11/24/2025) | \$92.44             |
| Kennametal Inc. (KMT.N)                          | E (01/08/2024) | \$27.85             |
| Lincoln Electric Holdings Inc (LECO.O)           | U (01/08/2024) | \$245.41            |
| Martin Marietta Materials Inc (MLM.N)            | O (01/08/2024) | \$624.37            |
| Oshkosh Corp (OSK.N)                             | E (01/08/2024) | \$129.43            |
| PACCAR Inc (PCAR.O)                              | E (04/16/2025) | \$110.33            |
| REV Group Inc. (REVG.N)                          | E (06/18/2025) | \$56.39             |
| Terex Corp. (TEX.N)                              | E (11/04/2025) | \$49.83             |
| Timken Co (TKR.N)                                | O (12/17/2024) | \$83.21             |
| United Rentals Inc. (URI.N)                      | O (04/16/2025) | \$796.91            |
| Vulcan Materials Company (VMC.N)                 | E (01/08/2024) | \$294.87            |
| Westinghouse Air Brake Technologies Corp (WAB.N) | O (01/08/2024) | \$214.95            |
| WillScot Holdings Corporation (WSC.O)            | E (11/12/2025) | \$20.61             |

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\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Semiconductors

| COMPANY (TICKER)                      | RATING (AS OF) | PRICE* (12/05/2025) |
|---------------------------------------|----------------|---------------------|
| <b>Joseph Moore</b>                   |                |                     |
| Advanced Micro Devices (AMD.O)        | E (06/09/2024) | \$217.97            |
| Aeva Technologies Inc (AEVA.O)        | E (07/19/2021) | \$14.70             |
| Allegro Microsystems Inc (ALGM.O)     | E (05/09/2025) | \$29.22             |
| Ambarella Inc (AMBA.O)                | O (03/29/2016) | \$74.10             |
| Amkor Technology Inc (AMKR.O)         | E (11/08/2023) | \$43.23             |
| Analog Devices Inc. (ADI.O)           | O (11/16/2023) | \$281.29            |
| Astera Labs Inc (ALAB.O)              | O (05/11/2025) | \$161.23            |
| Broadcom Inc. (AVGO.O)                | O (06/09/2024) | \$390.24            |
| GlobalFoundries Inc (GFS.O)           | E (10/28/2024) | \$38.95             |
| Intel Corporation (INTC.O)            | E (02/22/2023) | \$41.41             |
| IonQ Inc (IONQ.N)                     | E (04/25/2023) | \$52.69             |
| Marvell Technology Group Ltd (MRVL.O) | E (09/14/2015) | \$98.91             |
| Microchip Technology Inc. (MCHP.O)    | E (07/10/2024) | \$65.81             |
| Micron Technology Inc. (MU.O)         | O (10/06/2025) | \$237.22            |
| Navitas Semiconductor Corp (NVT.S.O)  | U (04/06/2025) | \$9.48              |
| NVIDIA Corp. (NVDA.O)                 | O (03/16/2023) | \$182.41            |
| NXP Semiconductor NV (NXPI.O)         | O (02/11/2025) | \$227.95            |
| ON Semiconductor Corp. (ON.O)         | E (05/11/2025) | \$54.74             |
| Qorvo Inc (QRVO.O)                    | E (10/28/2025) | \$89.96             |
| Qualcomm Inc. (QCOM.O)                | ++             | \$174.81            |
| SanDisk Corporation. (SNDK.O)         | O (03/03/2025) | \$228.47            |
| Semtech Corp. (SMT.C.O)               | E (04/06/2025) | \$79.61             |

|                                     |                 |          |
|-------------------------------------|-----------------|----------|
| Silicon Laboratories Inc. (SLAB.O)  | E (01/19/2021)  | \$141.52 |
| Skyworks Solutions Inc (SWKS.O)     | E (11/28/2018)  | \$69.32  |
| Texas Instruments (TXN.O)           | U (04/13/2020)  | \$182.54 |
| Wolfspeed, INC (WOLF.N)             | NR (04/06/2025) | \$22.07  |
| <b>Lee Simpson</b>                  |                 |          |
| Arm Holdings plc (ARM.O)            | O (07/19/2024)  | \$141.31 |
| Cadence Design Systems Inc (CDNS.O) | O (02/14/2024)  | \$337.53 |
| Synopsys Inc. (SNPS.O)              | O (11/10/2023)  | \$466.76 |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Multi-Industry

| COMPANY (TICKER)                           | RATING (AS OF) | PRICE* (12/05/2025) |
|--------------------------------------------|----------------|---------------------|
| <b>Christopher Snyder, CFA</b>             |                |                     |
| 3M Co. (MMM.N)                             | E (10/21/2025) | \$167.48            |
| Acuity Inc. (AYI.N)                        | O (01/14/2025) | \$371.99            |
| Allegion Public Limited Company (ALLE.N)   | E (01/06/2025) | \$159.08            |
| Alliance Laundry Holdings Inc (ALH.N)      | E (11/03/2025) | \$22.17             |
| Ametek Inc. (AME.N)                        | E (04/14/2025) | \$199.80            |
| Carrier Global Corp. (CARR.N)              | E (09/06/2024) | \$54.29             |
| Dover Corp (DOV.N)                         | E (04/14/2025) | \$191.09            |
| Eaton Corporation PLC (ETN.N)              | O (09/06/2024) | \$337.66            |
| Emerson Electric Co (EMR.N)                | U (09/06/2024) | \$137.39            |
| Fastenal Co. (FAST.O)                      | E (09/06/2024) | \$41.50             |
| Fortive Corp (FTV.N)                       | E (09/02/2025) | \$53.75             |
| Gates Industrial Corporation PLC (GTES.N)  | E (09/06/2024) | \$21.84             |
| Honeywell International Inc (HON.O)        | E (09/06/2024) | \$191.33            |
| Hubbell Inc. (HUBB.N)                      | E (09/06/2024) | \$440.53            |
| Ingersoll Rand INC (IR.N)                  | E (09/06/2024) | \$79.86             |
| Johnson Controls International Plc (JCI.N) | O (09/06/2024) | \$114.23            |
| Lennox International Inc (LII.N)           | U (01/06/2025) | \$511.63            |
| Otis Worldwide Corp (OTIS.N)               | E (09/06/2024) | \$86.45             |
| Parker-Hannifin Corp (PH.N)                | E (06/05/2025) | \$880.00            |
| QXO Inc (QXO.N)                            | O (09/02/2025) | \$20.82             |
| Ralliant Corporation (RAL.N)               | O (09/02/2025) | \$51.10             |
| Rockwell Automation (ROK.N)                | O (09/06/2024) | \$404.29            |
| Stanley Black & Decker Inc (SWK.N)         | E (09/06/2024) | \$72.47             |
| Trane Technologies PLC (TT.N)              | O (09/06/2024) | \$401.43            |
| Vertiv Holdings Co. (VRT.N)                | O (01/06/2025) | \$189.02            |
| W.W. Grainger Inc. (GWW.N)                 | E (09/06/2024) | \$975.54            |
| Watsco Inc (WSO.N)                         | E (04/14/2025) | \$347.77            |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Internet

| COMPANY (TICKER)              | RATING (AS OF) | PRICE* (12/05/2025) |
|-------------------------------|----------------|---------------------|
| <b>Brian Nowak, CFA</b>       |                |                     |
| Airbnb Inc (ABNB.O)           | U (12/06/2022) | \$124.32            |
| Alphabet Inc. (GOOGL.O)       | O (08/11/2015) | \$321.27            |
| Amazon.com Inc (AMZN.O)       | O (04/24/2015) | \$229.53            |
| Booking Holdings Inc (BKNG.O) | E (01/09/2019) | \$5,197.04          |
| DoorDash Inc (DASH.O)         | O (02/21/2024) | \$225.00            |
| Expedia Inc. (EXPE.O)         | E (01/09/2019) | \$263.43            |



|                                           |                |          |
|-------------------------------------------|----------------|----------|
| Instacart (CART.O)                        | E (01/29/2024) | \$44.18  |
| Lyft Inc (LYFT.O)                         | E (10/24/2019) | \$22.97  |
| Meta Platforms Inc (META.O)               | O (03/20/2023) | \$673.42 |
| Nextdoor Holdings Inc (NXDR.N)            | U (05/12/2025) | \$1.95   |
| Pinterest Inc (PINS.N)                    | O (07/20/2025) | \$26.75  |
| Reddit Inc (RDDT.N)                       | O (12/08/2024) | \$234.11 |
| Snap Inc. (SNAP.N)                        | E (07/22/2024) | \$7.91   |
| Uber Technologies Inc (UBER.N)            | O (06/04/2019) | \$91.32  |
| <b>Matthew Cost</b>                       |                |          |
| AppLovin Corp (APP.O)                     | O (04/10/2025) | \$691.94 |
| Compass, Inc. (COMP.N)                    | ++             | \$10.32  |
| Criteo SA (CRO.O)                         | E (01/26/2016) | \$21.00  |
| DoubleVerify Holdings Inc (DV.N)          | E (06/25/2024) | \$11.03  |
| Electronic Arts Inc (EA.O)                | E (08/04/2021) | \$203.92 |
| Integral Ad Science Holding Corp. (IAS.O) | E (04/16/2024) | \$10.28  |
| MNTN Inc (MNTN.N)                         | E (06/16/2025) | \$13.05  |
| Opendoor Technologies Inc (OPEN.O)        | E (07/24/2023) | \$7.15   |
| Playtika Holding Corp (PLTK.O)            | E (11/27/2022) | \$4.24   |
| Roblox Corporation (RBLX.N)               | O (11/04/2024) | \$96.28  |
| Shutterstock Inc (SSTK.N)                 | E (07/28/2022) | \$19.01  |
| Take-Two Interactive Software (TTWO.O)    | O (02/01/2018) | \$247.88 |
| Trade Desk Inc (TTD.O)                    | E (09/10/2025) | \$40.05  |
| Unity Software Inc (U.N)                  | O (09/02/2024) | \$45.78  |
| Webtoon Entertainment Inc (WBTN.O)        | E (07/22/2024) | \$13.91  |
| Yelp Inc (YELP.N)                         | U (01/10/2019) | \$29.68  |
| Zillow Group Inc (Z.O)                    | E (04/18/2018) | \$74.04  |
| <b>Nathan Feather</b>                     |                |          |
| Bumble Inc. (BMBL.O)                      | E (03/08/2021) | \$3.62   |
| Chewy Inc (CHWY.N)                        | O (10/31/2023) | \$33.47  |
| Duolingo Inc (DUOL.O)                     | O (04/23/2025) | \$199.75 |
| eBay Inc (EBAY.O)                         | O (04/18/2024) | \$82.47  |
| Etsy Inc (ETSY.N)                         | E (07/20/2025) | \$53.26  |
| FIGS, Inc. (FIGS.N)                       | E (02/29/2024) | \$11.43  |
| Match Group Inc (MTCH.O)                  | E (04/18/2024) | \$34.52  |
| Peloton Interactive, Inc. (PTON.O)        | E (03/14/2022) | \$6.42   |
| Revolve Group Inc (RVLV.N)                | E (10/20/2024) | \$25.97  |
| WW International Inc (WW.O)               | E (08/01/2025) | \$24.11  |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.